

Boston Infinite Money Glitch:

Hacking Transit Cards Without Ending Up In Handcuffs

Matthew Harris, Zachary Bertocchi, Scott Campbell, Noah Gibson

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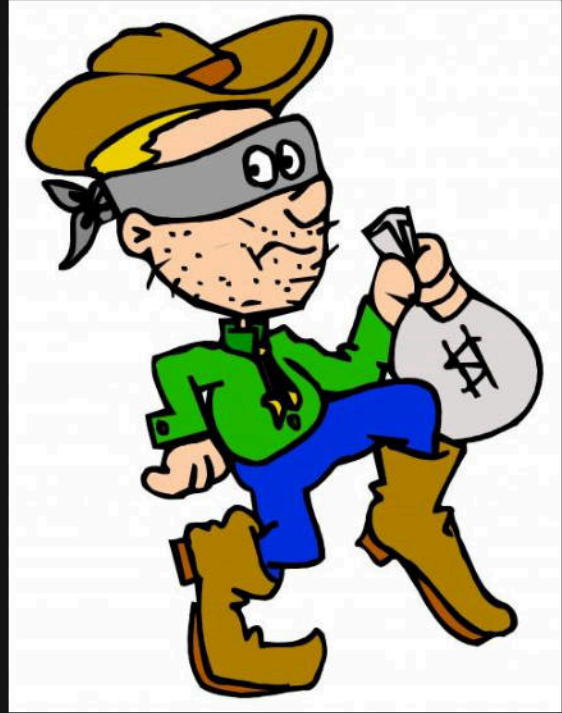
Matthew Harris, Zachary Bertocchi, Scott Campbell, Noah
Gibson

* The MBTA's lawyers have kindly asked us to say the following:

"The MBTA does not endorse or encourage this type of conduct. evading fares and/or hacking the MBTA fare system is illegal and the MBTA takes these matters seriously. Anyone caught engaging in this type of behavior will be referred to the appropriate law enforcement agency. The MBTA has implemented mitigation measures to address some of these concerns."

whoarewe

- Vocational high schoolers from Medford, MA
- ~~Professional~~ Amateur miscreants
- People who don't like being told what to do
- Underwater robotics enjoyers





Matthew Harris



Zack Bertocchi



Scott Campbell



Noah Gibson

“What am I getting out of this?”

- **Free rides on the MBTA**
- **A lesson on how to reverse engineer a transit card**
- **A lesson on how to not end up in handcuffs for the above two items**
- **A fun story, hopefully**

THIS IS ILLEGAL

(don't do it)

(or at least don't tell us about it)

Introduction

Meet the T

- Massachusetts Bay Transportation Authority - The public transit agency in Boston
- First transit system in the country



Image Credit: MBTA



Image Credit: MBTA

Sounds great! How can I hop on?

Option 1: Traditional fare evasion

Sounds great! How can I hop on?

Option 1: Traditional fare evasion

- Hop the gates



Sounds great! How can I hop on?

Option 1: Traditional fare evasion

- Hop the gates
- Force the gates open



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- Hop the gates
- Force the gates open
- Too hard to explain, just look



Sounds great! How can I hop on?

Option 1: Traditional fare evasion

- Hop the gates
- Force the gates open
- Too hard to explain, just look



This works too, I guess

Option 2: Pay

- Subway fare - \$2.40
- Bus fare - \$1.70



The MBTA has authorized us to use the CharlieCard and CharlieTicket logos

The Problem With These Strategies

Traditional Fare Evasion

- Too easy
- Boring
- Might get caught

Paying

- Too legal
- Way too boring
- Costs money

What to do?

Hack the fare system

Hack the fare system

duh

Part 1: Humble Beginnings



WIKIPEDIA
The Free Encyclopedia

Main page
Contents
Current events
Random article
About Wikipedia
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What links here
Related changes
Special pages
Permanent link
Page information
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Languages
Français
中文

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Article Talk

CharlieCard

From Wikipedia, the free encyclopedia

The **CharlieCard** is a *contactless smart card* used for fare payment for transportation in the Boston area. It is the primary payment method for the Massachusetts Bay Transportation Authority (MBTA) and several regional public transport systems in the U.S. state of Massachusetts.

The card was introduced on December 4, 2006, to enhance the technology of the transit system and eliminate the burden of carrying and collecting tokens.^[3] It replaces the metal token, the last one of which was sold at Government Center station on December 6, 2006.^[4] It is named after a fictional character in the *folk* song "M.T.A.", often called "Charlie on the MTA", which concerns a man forever trapped on the Boston subway system – then known as the Metropolitan Transit Authority (MTA) – because he cannot pay the 5-cent *surcharge required to leave the train*.

In 2022, the original CharlieCard system will be replaced during the "Automated Fare Collection 2.0" project, a system similar to the London Oyster Card. The new system will allow payments with contactless cards and smartphones, as well as new CharlieCards.^[5]

Contents [hide]

- History
- Technology
 - Smartphone technology
 - Effect on transit employees
- Card types
 - CharlieTicket
 - Bike CharlieCard
- Purchase options
- Future
 - Automated Fare Collection 2.0
- Criticism
 - Green Line inefficiency
 - Security concerns
- See also
- References
- Further reading
- External links

History [edit]

The CharlieCard is named after the lead character in the 1948 protest *folk music* song, "M.T.A.". The song was written to protest a fare increase in the form of an extra five cent *exit fare* for longer rides and was later made popular by the Kingston Trio in 1959.^{[6][7]} One of the rejected names for the farecard system was "The Fare Cod", a pun on both the way locals might pronounce "Card" and the fish that was once integral to the Massachusetts economy, and also a reference to other transit cards named for ocean animals, such as London's Oyster and Hong Kong's Octopus. Another rejected name was T Go card with the T being the symbol for the MBTA.^[8]

CharlieCards work on the MBTA's subway and bus services, most of which were converted in 2006. Token sales ended on December 6, 2006.^[9] The final fare-controlled station to be converted was Fields Corner station on December 22, 2006.^[10] They were originally expected to be usable on MBTA commuter rail and ferry boat services by December 2008,^[11] with testing on the Commuter Rail originally planned for summer 2008.^[12] However,

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Read Edit View history



CharlieCard



Location Boston, Massachusetts, U.S.

Launched December 4, 2006

Technology Contactless smart card
MIFARE

Manager Massachusetts Bay
Transportation Authority

Currency USD

Validity Massachusetts Bay
Transportation Authority
Berkshire Regional Transit
Authority
Brockton Area Transit Authority
Cape Ann Transportation
Authority
Cape Cod Regional Transit
Authority
Lowell Regional Transit Authority
Merrimack Valley Regional
Transit Authority
MetroWest Regional Transit
Authority (Discontinued support
for CharlieCard since May 2,
2022, due to new fare system.
New primary method for payment
on MWRTA is named "Catch
Card" ^[13])
Montachusett Regional Transit
Authority (Discontinued support
for CharlieCard since June 1,
2022, due to new fare system.
New primary method for payment

Security concerns [\[edit \]](#)

See also: *Massachusetts Bay Transportation Authority v. Anderson*

Security flaws in the CharlieCard technology were studied and reported in a presentation by Henryk Plötz and Karsten Nohl at the [Chaos Communication Congress](#) in December 2007, which described a partial reverse-engineering of the algorithm used in the MIFARE Classic chip.^[71] The MIFARE Classic smartcard^[72] from [NXP Semiconductors](#), owned by [Philips](#), was reported as compromised in March 2008 by a group of researchers led by Karsten Nohl, a Ph.D. student in the Department of Computer Science, [University of Virginia](#).^{[73][74][75]}

In addition, the security used on the mag-stripe CharlieTickets was broken by a team of [MIT](#) students. They were scheduled to give a talk about their findings at [DEFCON 16](#) in August 2008,^[76] but were stopped after a [federal lawsuit](#) was filed against them by the MBTA, which resulted in a restraining order being issued.^{[77][78]} However, their presentation had already been published by DEFCON before the complaint was filed.^[79] On August 19, the court ruled the students could give their presentation.^[80]

Other MIT students leveraged the technology behind Charlie Cards in 2013, with the development of Sesame Ring, a wearable ring embedded with an RFID tag that would save riders time in passing through MBTA station faregates.^[81] The students formed a company called Ring Theory and funded development of the product using a [Kickstarter](#) campaign. The Sesame Ring can be ordered online, or purchased in the MBTA Gift Store in Cambridge.^[82] The product was developed with full cooperation from the MBTA.^[83]

In 2022, it was revealed that the NFC chip in some Android smartphones could interact with CharlieCards, including duplicating data from one card to a blank card. The MBTA indicated that its software systems detected a small number of such duplicated cards – about ten per month – which were then deactivated.^[84]

The OGs

- 15 years ago
- Some MIT students found a vulnerability in the CharlieCards and Tickets
- Figured out how to clone and forge CharlieTickets
- Wanted to present at DEF CON
- How'd they get thanked for their research?



UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

MASSACHUSETTS BAY
TRANSPORTATION AUTHORITY

Plaintiff

v.

ZACK ANDERSON, RJ RYAN,
ALESSANDRO CHIESA, and the
MASSACHUSETTS INSTITUTE OF
TECHNOLOGY

Defendants

Civil Action No. No. 08- 11364-GAO

DECLARATION OF SCOTT HENDERSON

1. I am the Systems Project Manager for the Automated Fare Collection System for the plaintiff, Massachusetts Bay Transportation Authority ("MBTA").
2. My duties include (a) responsibility for the computer systems resident in all of the devices within the Automated Fare Collection ("AFC") System, and (b) responsibility for all software application systems within the AFC System. I have included further information concerning my qualifications, relevant to this Declaration, in paragraphs 28 through 37.
3. I make this declaration based on my personal knowledge and a review of MBTA business records concerning the matters set out below.

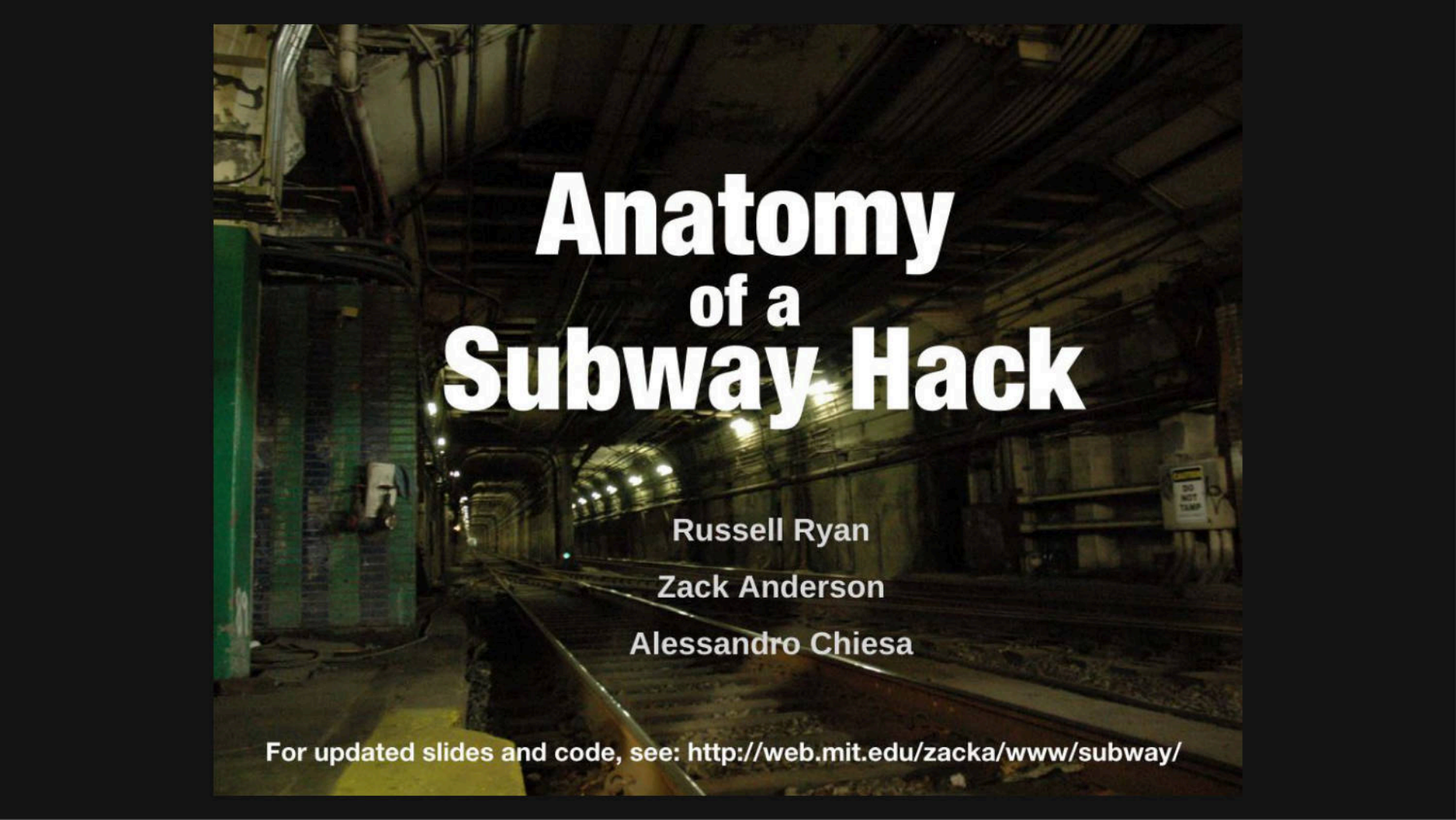
Requests To The MIT Undergrads For Their DEFCON Presentation Materials

4. I have reviewed the documents referred to in the Complaint as the " Initial Announcement" and the "Revised Announcement". I use the term "MIT Undergrads" to refer to the defendants, Zack Anderson, RJ Ryan, and Alessandro Chiesa.

DefCon: Boston Subway Officials Sue to Stop Talk on Fare Card Hacks — Update: Restraining Order Issued; Talk Cancelled

BY KIM ZETTER 08.08.08 | 11:45 PM | [PERMALINK](#)





Anatomy of a Subway Hack

Russell Ryan

Zack Anderson

Alessandro Chiesa

For updated slides and code, see: <http://web.mit.edu/zacka/www/subway/>

Let's try to be like them

Maybe minus the lawsuit part

Step 1: Go for the low hanging fruit

- CharlieTickets have awful security
- MIT students found how to clone and reverse engineer them
- Seems easy enough
- Free rides 4 life!



Step 2: Choose your weapon

- **MSR206**
- **88 bucks**
- **A bit pricey, but it's an investment**
- **Should we go ahead and buy it?**



Click image to open expanded view

(MSR605 & 206 Magnetic Card Reader & Program Software for Windows 98/Me/XP/Vista/Windows7 & 3-track version can read/write all three tracks data, 300-4000 oe & Support USB Communication) MSR605 Magnetic Card Reader Writer Encoder Stripe Swipe Credit Magstripe MSR206

[Visit the Bravolink Store](#)

4.1 ★★★★★ 177 ratings | 90 answered questions

\$88⁰⁰

[FREE Returns](#)

Color	Black
Compatible Devices	Printer
Data Transfer Rate	480 Megabits Per Second
Operating System	Windows 98
Hardware Interface	USB

Maybe not

“We’re phasing out the old CharlieTickets for new, tappable versions. Riders can purchase the new CharlieTickets at new fare vending machines across the system.”



September 2021

Time for a new plan



Meet the CharlieCard

- Contactless smart card
- Introduced in 2004
- Add money at vending machines
- Tap cards at fare gates



But how does it work?

- Not 100% sure yet
- We know it uses NFC
- Probably uses MIFARE under the hood

MIFARE?

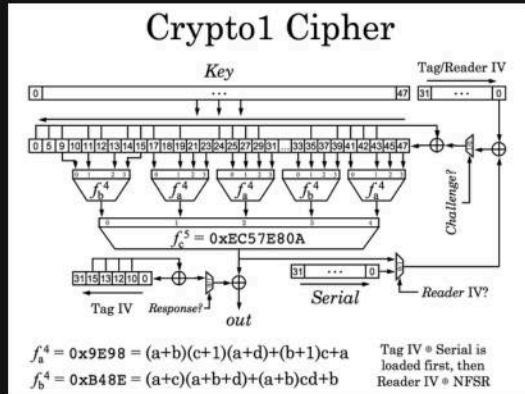
- **Standard for data storage and communication**
- **Made by NXP**
- **Lots of different flavors of MIFARE**
- **Most infamous flavor: MIFARE Classic**



MIFARE Classic



- Oldest flavor
- Crypto-1 encryption algorithm: **Proprietary, 48-bit, secured through obscurity**
- History of lawsuits around it



How it's supposed to work



How it actually works



So how do we talk to the cards?

Step 1: Grab an NFC reader

- **ACR122U**
- **\$40**
- **USB**
- **Software exists**



Roll over image to zoom in

[NFC RFID Reader Writer ACR 122U](#)
[ISO RFID Reader Writer ACR 14443A](#)
[/ B Free Software in White](#)
Brand: BITOUT

\$40⁴¹

Get \$60 off instantly: Pay **\$0.00** ~~\$40.41~~ upon approval for the Amazon Store Card. No annual fee.

- Supports ISO 14443 type A and B, FeliCa and all 4 types of NFC (ISO / IEC 18092) markings
- Integrated antenna for contactless tag access with card reading distance of up to 50 mm (depending on the day)
- Supports new Ultralight C (via pseudo APDUs) and Plus SL1 (4-byte UID over pseudo APDUS) and SL3

Step 2: Figure out how to use it

libnfc 1.7.1

[Main Page](#)[Modules](#)[Data Structures](#)[Files](#)[Directories](#)

libnfc reference manual

Introduction

This is the developer manual for **libnfc**. libnfc is an open source library that allows you to communicate with NFC devices. For more info, see the [libnfc homepage](#).

Quick start

If you are looking for libnfc's public API, you should start with the Modules page which links to the different categories of libnfc's functionality. Some commented examples that present how to use **libnfc** can be found here: [Examples](#)

Others example programs can be found in the libnfc source distribution under the "examples" subdirectory [examples](#).

You can also find utils in the libnfc source distribution under the "utils" subdirectory [utils](#).

Error handling

libnfc functions typically return 0 or more on success or a negative error code on failure. These negative error codes relate to LIBNFC_ERROR constants which are listed on the [Error reporting](#) documentation page.

Upgrading from previous version

If you are upgrading from a previous **libnfc** version, please take care about changes, specially API changes. All important changes should be listed in [ChangeLog](#).

Step 3: Send it back cause you don't know how to use it



Step 4: Buy another reader

- Proxmark3
- \$40-80
- Truly top-notch software
- Decently easy to use

SAVE UP TO 10% [See all eligible items and terms](#)

Specification: Model: PM3Easy3.0
Working voltage: 3.5-5.5V
Working current: 50-130mA
Length and width: 54mm*86.6mm
Color: Black
Thickness size:
6.2mm (the thinnest)
9.8mm (plus UF_ANT_500UH Portable ANT V3.0
Screw)
15.8 (plus low frequency antenna)



Proxmark3 Easy V3.0 RDV4 512k ID M1 RFID Antenna Card
Reader Integrated Antenna

★★★★★ Be the first to [write a review](#).

Condition: **New**

Quantity: More than 10 available / 70 sold

Price: **US \$46.72**
Was US \$56.38
Save US \$9.66 (17% off)

[Buy It Now](#)

[Add to cart](#)

Best Offer:

[Make offer](#)

[Add to watchlist](#)

This one's trending. 70 have already sold.

Breathe easy. Free shipping and returns.

Shipping: **Free Standard Shipping.** [See details](#)
Located in: Sacramento, California, United States

[Have one to sell?](#) [Sell now](#)

Wait a minute, it doesn't work!

Step 5: Buy another Proxmark



IC-M1

IC-UID

IC-CUID

ID-5577

**Proxmark3 Easy RFID Copier NFC RFID
Detector Duplicator Reader Writer Cloner
3.0 512K Memory with 2 USB IC ID 125 kHz
13.56 Mhz**

★★★★★ 5.0 2 Reviews 10 Sold

\$39.75 ~~\$46.76~~ 15% off

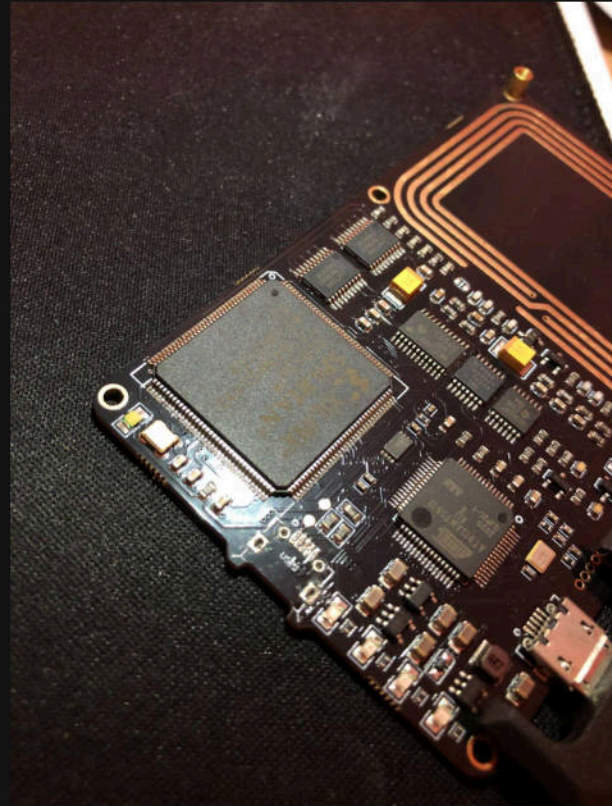
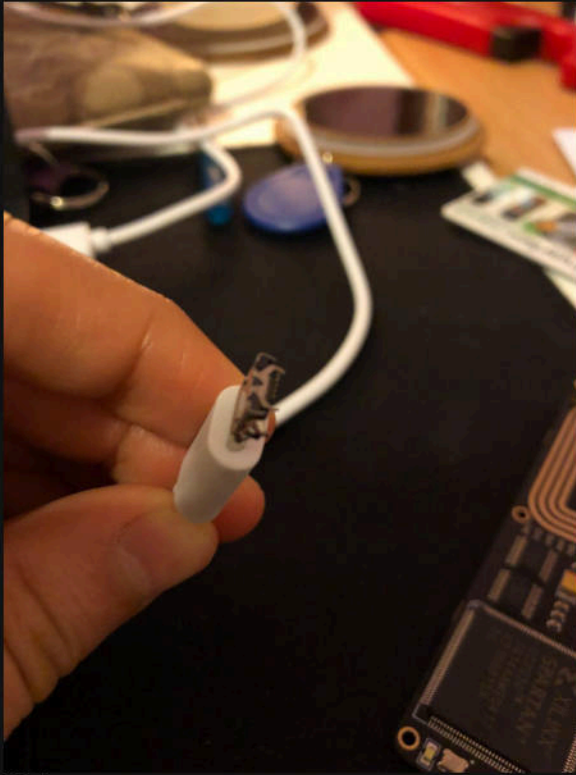
2% Off
Store Discount

\$1.00 Off
Store Coupon



It works!!!

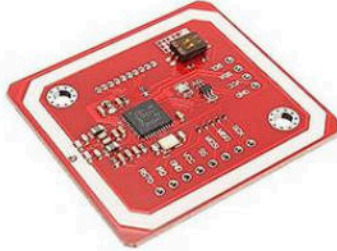
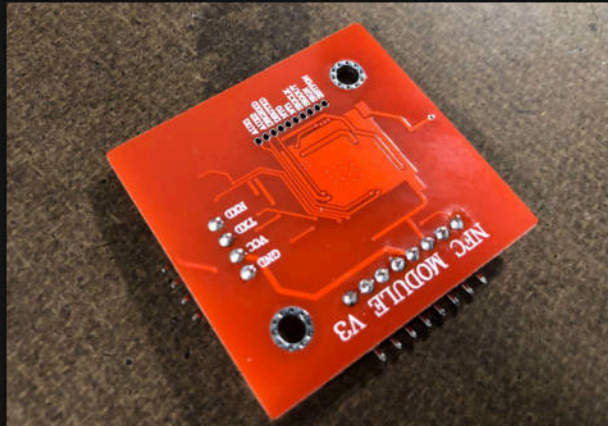
fuck



November 2021

Step ㄟ(ツ)ㄟ: Buy another reader

- PN532
- \$9
- Hope you like soldering
- Only works with Raspberry Pi



HiLetgo PN532 NFC NXP RFID Module V3 Kit Near Field Communication Reader Module Kit I2C SPI HSU with S50 White Card Key Card for Arduino Raspberry Pi DIY Smart Phone Android Phone

[Visit the HiLetgo Store](#)

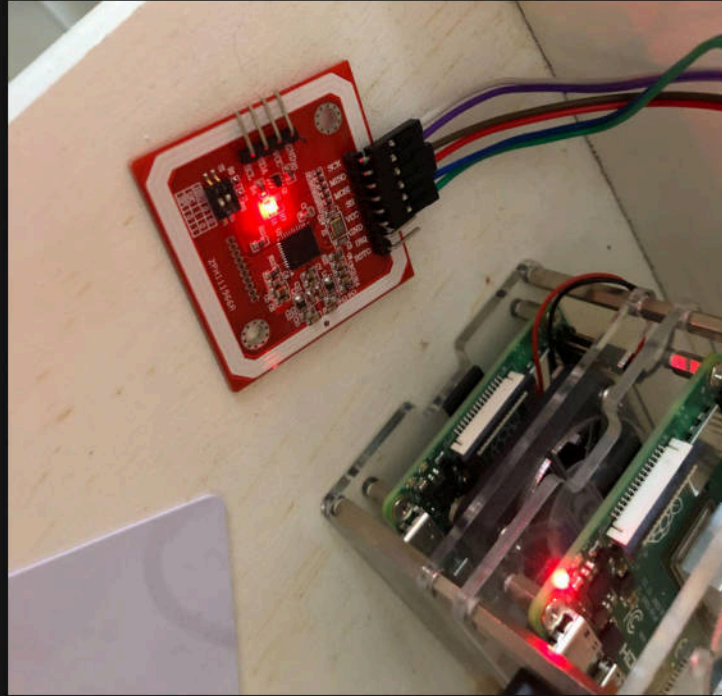
4.4 ★★★★★ 144 ratings | 9 answered questions

Amazon's Choice for "pn532"

Roll over image to zoom in

\$8⁹⁹

It works! For real this time



Next Step: Get the keys

- Bad encryption is still encryption
- Still need keys to read and write
- How do we get them?



Looks promising



black hat
REGIONAL SUMMIT
SAO PAULO 2014

Hacking Mifare Classic Cards

Márcio Almeida (marcioalma@gmail.com)

01010100100010110111100110101101010010111010110101011111011010011101001011010
101101010010101010101101101010010101010110110101001010101011011010100101
110110101001010101010110110101001010101011011010100101010101101101010010
1010100100010110111100110101101010010111010110101011111011010011101001011010
101101010010101010101101101010010101010110110101001010101011011010100101
110110101001010101010110110101001010101011011010100101010101101101010010

- Data

- 
- blackhat**

[illegible]

Let's give them a try!

mfoc

- **Implements the nested attack**
- **Does some fancy magic to find a card's keys**
- **Let's try it!**

mfoc

- **Implements the nested attack**
- **Does some fancy magic to find a card's keys**
- **Let's try it!**

mloc

- Implements the nested attack
- Does some fancy magic to find a card's keys
- Let's try it!
- Requires an initial key

```
Try to authenticate to all sectors with default keys...
Symbols: '.' no key found, '/' A key found, '\' B key found, 'x' both keys found
[Key: ffffffff] -> [.....]
[Key: a0a1a2a3a4a5] -> [.....]
[Key: d3f7d3f7d3f7] -> [.....]
[Key: 000000000000] -> [.....]
[Key: b0b1b2b3b4b5] -> [.....]
[Key: 4d3a99c351dd] -> [.....]
[Key: 1a982c7e459a] -> [.....]
[Key: aabbccddeeff] -> [.....]
[Key: 714c5c886e97] -> [.....]
[Key: 587ee5f9350f] -> [.....]
[Key: a0478cc39091] -> [.....]
[Key: 533cb6c723f6] -> [.....]
[Key: 8fd0a4f256e9] -> [.....]
```

```
Sector 00 - Unknown Key A      Unknown Key B
Sector 01 - Unknown Key A      Unknown Key B
Sector 02 - Unknown Key A      Unknown Key B
Sector 03 - Unknown Key A      Unknown Key B
Sector 04 - Unknown Key A      Unknown Key B
Sector 05 - Unknown Key A      Unknown Key B
Sector 06 - Unknown Key A      Unknown Key B
Sector 07 - Unknown Key A      Unknown Key B
Sector 08 - Unknown Key A      Unknown Key B
Sector 09 - Unknown Key A      Unknown Key B
Sector 10 - Unknown Key A      Unknown Key B
Sector 11 - Unknown Key A      Unknown Key B
Sector 12 - Unknown Key A      Unknown Key B
Sector 13 - Unknown Key A      Unknown Key B
Sector 14 - Unknown Key A      Unknown Key B
Sector 15 - Unknown Key A      Unknown Key B
```

mloc: ERROR:

No sector encrypted with the default key has been found, exiting..

mfcuk

- Implements the dark side attack
- Finds the keys from out of nowhere
- Let's try it!

```
mfcuk - 0.3.8
Mifare Classic DarkSide Key Recovery Tool - 0.3
by Andrei Costin, zveriu@gmail.com, http://andreicostin.com

WARN: cannot open template file './data/tmpls_fingerprints/mfcuk_tmpl_skgd.mfd'
WARN: cannot open template file './data/tmpls_fingerprints/mfcuk_tmpl_ratb.mfd'
WARN: cannot open template file './data/tmpls_fingerprints/mfcuk_tmpl_oyster.mfd'

INFO: Connected to NFC reader: ACS / ACR122U PICC Interface

VERIFY:
    Key A sectors: 0 1 2 3 4 5 6 7 8 9 a b c d e f
    Key B sectors: 0 1 2 3 4 5 6 7 8 9 a b c d e f

RECOVER: 0
```



Last ditch effort



DuckDuckGo

MIFARE Classic default keys

mifare classic **tool key**

mifare classic **tool key file**

mifare classic **1k key fobs**

mifare classic **tools windows**

mifare-**key-cracker**

mifare **desfire master key**

mifare classic **1k driver**

mifare classic **tool windows download**

2177 lines (2176 sloc) | 29.1 KB

```
1  #
2  # Mifare Default Keys
3  #   -- iceman fork version --
4  #   -- contribute to this list, sharing is caring --
5  #
6  # Default key
7  FFFFFFFFFFFFFFFF
8  #
9  # Blank key
10 00000000000000
11 #
12 # NFC Forum MADkey
13 A0A1A2A3A4A5
14 #
```

holy shit

Sector 00 - Found	Key A: 3060206f5b0a Found	Key B: f1b9f5669cc8
Sector 01 - Found	Key A: 5ec39b022f2b Found	Key B: f662248e7e89
Sector 02 - Found	Key A: 5ec39b022f2b Found	Key B: f662248e7e89
Sector 03 - Found	Key A: 5ec39b022f2b Found	Key B: f662248e7e89
Sector 04 - Found	Key A: 5ec39b022f2b Found	Key B: f662248e7e89
Sector 05 - Found	Key A: 5ec39b022f2b Found	Key B: f66224ee1e89
Sector 06 - Found	Key A: 5ec39b022f2b Found	Key B: f66224ee1e89
Sector 07 - Found	Key A: 5ec39b022f2b Found	Key B: f66224ee1e89
Sector 08 - Found	Key A: 3a09594c8587 Found	Key B: 62387b8d250d
Sector 09 - Found	Key A: f238d78ff48f Found	Key B: 9dc282d46217
Sector 10 - Found	Key A: afd0ba94d624 Found	Key B: 92ee4dc87191
Sector 11 - Found	Key A: b35a0e4acc09 Found	Key B: 756ef55e2507
Sector 12 - Found	Key A: 447ab7fd5a6b Found	Key B: 932b9cb730ef
Sector 13 - Found	Key A: 1f1a0a111b5b Found	Key B: ad9e0a1ca2f7
Sector 14 - Found	Key A: d58023ba2bdc Found	Key B: 62ced42a6d87
Sector 15 - Found	Key A: 2548a443df28 Found	Key B: 2ed3b15e7c0f

We have all sectors encrypted with the default keys..

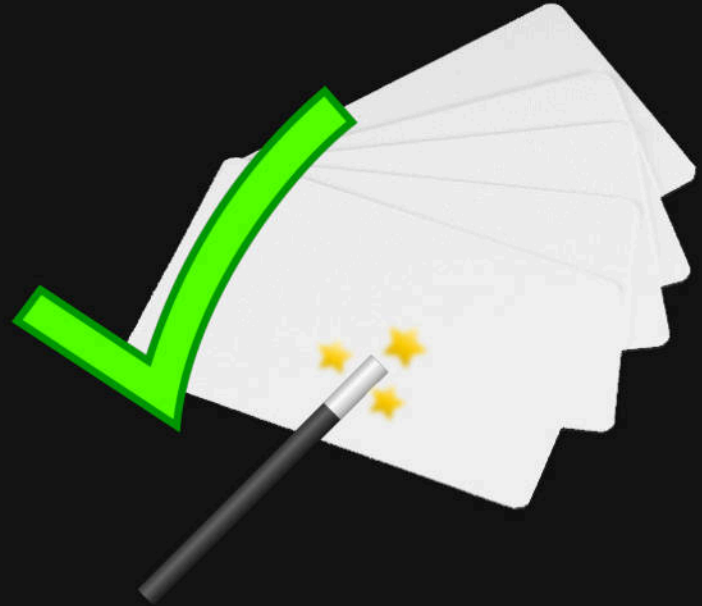
Let's grab a binary dump

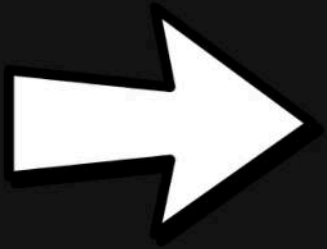
```
00000000 04 48 5A 35 23 88 04 00 C8 07 00 20 00 00 00 20 HZ5#.....
00000010 4E 0F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030 30 60 20 6F 5B 0A 78 77 88 C1 F1 B9 F5 66 9C C8 0` o[.xw.....f..
00000040 04 10 23 45 66 77 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000050 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00 00
00000060 00 20 20 00 00 00 00 00 00 00 00 00 00 00 00 00
00000070 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000080 11 9A 71 13 CF C0 94 B7 E9 65 00 0F 08 00 7F 0D .q.....e.....
00000090 5B FC 9E CD 00 00 FA 00 19 F3 4E 31 4C 78 52 E9 [.N1LxR.
000000A0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .>[
000000B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
000000C0 11 9A 71 13 CF C0 94 B7 E9 65 00 0E 80 00 8C FA .q.....e.....
000000D0 5B FC 9E CD 00 00 C8 00 19 F3 4E 31 4C 78 AB 33 [.N1Lx.3
000000E0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .>[
000000F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000100 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F .?
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F
00000120 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 48
00000130 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000140 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F .?
00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F
00000160 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 48
00000170 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$.~.
00000180 9A 82 B4 8F D8 80 C8 9A 82 B5 8F D8 80 32 D8 54 .....2.T
00000190 96 1C A0 60 B9 01 54 96 1C A8 60 B9 01 54 68 3F ....T....Th?
000001A0 96 65 D8 61 21 01 54 96 6B 49 13 D1 01 54 49 D6 .e.a!.T.kI...TI.
000001B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$.~.
000001C0 96 6B 56 62 49 00 00 96 AE C4 23 40 80 33 03 F5 .kVbI.....#@.3..
000001D0 9A 71 0F 8F C8 80 D2 9A 71 13 CF C1 01 E0 4C DE .q.....q.....L.
000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F
000001F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$.~.
00000200 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000210 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000220 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000230 3A 09 59 4C 85 87 78 77 88 00 62 38 7B 8D 25 0D .:YL.xw...b8{.%.
00000240 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000250 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000260 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000270 F2 38 D7 8F F4 8F 78 77 88 00 9D C2 82 D4 62 17 .8....xw.....b.
```

Part 2: Cloning Cards

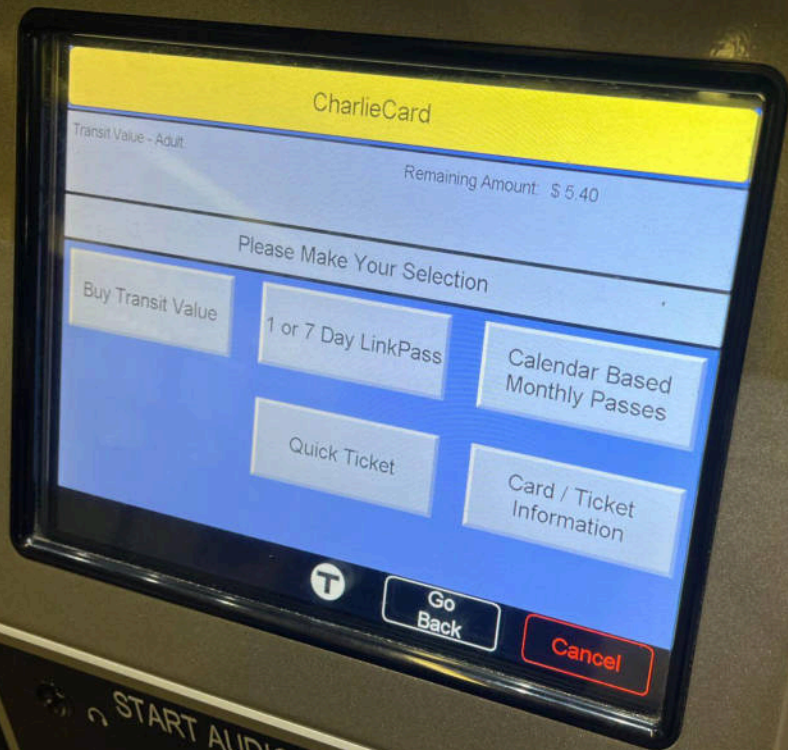
November 2021

Magic? I love magic!!!





A hand holding a black pen with a silver tip, pointing at a stack of white cards. The top card features three yellow stars arranged in a triangular pattern. The cards are slightly fanned out, and the background is black.



START AUDIO PRESS 5 >

Wait wait... where's the source of truth here?

The card

Let's automate it!



```
nvim main.c
main.c
341 }
342
343 static void print_usage() {
344     printf("Usage: ");
345     printf("charlie-clone [options] <file>\n");
346     printf("  -h, --help    Show this help page\n");
347     printf("  -d, --dump    Save the contents of the card to a file\n");
348     printf("  -f, --file    Read the contents of a file instead of a card,"
349             "and write them to a blank card\n");
350
351     printf("Examples: \n");
352     printf("  charlie-clone -f card_data.mfd    Read the data from the file\n");
353     printf("  charlie-clone -f card_data.mfd    Read the data from the file and write it to a blank card\n");
354     printf("  charlie-clone -d card_data.mfd    Save the contents of the card\n");
355     printf("  charlie-clone -f card_data.mfd\n");
356 }
357
358 int main(int argc, char **argv) {
359     int option;
360     int option_index;
361     bool dumping_to_file = false;
362     char *dump_file_name = "";
363     bool reading_from_file = false;
364     char *read_file_name = "";
365
366     static struct option long_options[] = {{ "dump", required_argument, 0, 'd'},
367                                           { "file", required_argument, 0, 'f'},
368                                           { "help", no_argument, 0, 'h'},
369                                           { 0, 0, 0, 0}};
370
371     if (argc > 3) {
372         printf("Error: Too many arguments!\n");
373         print_usage();
374         exit(1);
375     }
376
377     while (true) {
378         option = getopt_long(argc, argv, "hd:f:", long_options, &option_index);
379         if (option == -1) {
380             break;
381         }
382         switch (option) {
383             case 'h':
384                 print_usage();
385                 exit(0);
386             case 'd':
387                 dumping_to_file = true;
388                 dump_file_name = optarg;
389                 break;
390             case 'f':
391                 reading_from_file = true;
392                 read_file_name = optarg;
393                 break;
394         }
395     }
396 }
```

```
~/Projects/charlie-clone/src charlie-clone
charlie-clone v1.0.0
```

```
NFC reader: ACS / ACR122U PICC Interface opened
Place the CharlieCard on the reader!
Card found!
Found CharlieCard:
ISO/IEC 14443A (106 kbps) target:
  ATQA (SENS_RES): 00 04
  UID (NFCID1): f4 22 89 34
  SAK (SEL_RES): 08
```

```
[#####] 100%
```

```
Please remove card from reader
Card removed
```

```
Please place a blank magic card on the reader
Card found!
```

```
Found magic card:
ATQA (SENS_RES): 00 04
UID (NFCID1): f4 22 89 34
SAK (SEL_RES): 88
```

```
Card unlocked!
[#####] 100%
```

```
~/Projects/charlie-clone/src
```


Infinite Money??? Not so fast...

- Requires upfront investment
- Might set off some fraud alarms
- When your card gets disabled you need to PAY to buy another one



Cloning Cards is Cool But Y'know What's Cooler?

Forging cards



Part 3: Cracking the Code

November 2021

Reverse Engineering a CharlieCard: Where do we start?

Let's take a look at the hex dump!

```
00000000 04 48 5A 35 23 88 04 00 C8 07 00 20 00 00 00 20 HZ5#.....
00000010 4E 0F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030 30 60 20 6F 5B 0A 78 77 88 C1 F1 B9 F5 66 9C C8 0 o[.xw....f..
00000040 04 10 23 45 66 77 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000050 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00 00
00000060 00 20 20 00 00 00 00 00 00 00 00 00 00 00 00 00
00000070 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000080 11 9A 71 13 CF C0 94 B7 E9 65 00 0F 08 00 7F 0D ..q.....e....
00000090 5B FC 9E CD 00 00 FA 00 19 F3 4E 31 4C 78 52 E9 [.....N1LxR.
000000A0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .....>[
000000B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
000000C0 11 9A 71 13 CF C0 94 B7 E9 65 00 0E 80 00 8C FA ..q.....e....
000000D0 5B FC 9E CD 00 00 C8 00 19 F3 4E 31 4C 78 A0 33 [.....N1Lx.3
000000E0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .....>[
000000F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000100 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F .....?
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
00000120 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B .....K
00000130 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000140 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F .....?
00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
00000160 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B .....K
00000170 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
00000180 9A 82 B4 BF D8 80 C8 9A 82 B5 BF D8 80 32 D8 54 .....2.T
00000190 96 1C A0 60 B9 01 54 96 1C A8 60 B9 01 54 68 3F .....T.....Th?
000001A0 96 65 D8 61 21 01 54 96 6B 49 13 D1 01 54 49 D6 ..e.a!.T.kI...TI.
000001B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
000001C0 96 6B 56 62 49 00 00 96 AE C4 23 40 80 33 03 F5 ..kVbI.....#@.3..
000001D0 9A 71 0F BF C8 80 D2 9A 71 13 CF C1 01 E0 4C DE ..q.....q.....L.
000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
000001F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
00000200 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000210 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000220 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000230 3A 09 59 4C 85 87 78 77 88 00 62 38 7B 8D 25 0D ..YL...xw...b8{.%.
00000240 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000250 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000260 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000270 F2 38 D7 8F F4 8F 78 77 88 00 9D C2 82 D4 62 17 ..8....xw.....b.
```

Looks like the serial number

```
00000000 04 48 5A 35 23 88 04 00 C8 07 00 20 00 00 00 20 HZ5#.....
00000010 4E 0F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030 30 60 20 6F 5B 0A 78 77 88 C1 F1 B9 F5 66 9C C8 0 o[.xw....f..
00000040 04 10 23 45 66 77 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000050 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00
00000060 00 20 20 00 00 00 00 00 00 00 00 00 00 00 00
00000070 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000080 11 9A 71 13 CF C0 94 B7 E9 65 00 0F 08 00 7F 0D ..q.....e....
00000090 5B FC 9E CD 00 00 FA 00 19 F3 4E 31 4C 78 52 E9 [.....N1LxR.
000000A0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .....>[
000000B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
000000C0 11 9A 71 13 CF C0 94 B7 E9 65 00 0E 80 00 8C FA ..q.....e....
000000D0 5B FC 9E CD 00 00 C8 00 19 F3 4E 31 4C 78 A0 33 [.....N1Lx.3
000000E0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .....>[
000000F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000100 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F .....?
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
00000120 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B .....K
00000130 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000140 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F .....?
00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
00000160 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B .....K
00000170 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
00000180 9A 82 B4 BF D8 80 C8 9A 82 B5 BF D8 80 32 D8 54 .....2.T
00000190 96 1C A0 60 B9 01 54 96 1C A8 60 B9 01 54 68 3F .....T.....Th?
000001A0 96 65 D8 61 21 01 54 96 6B 49 13 D1 01 54 49 D6 ..e.a!.T.kI...TI.
000001B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
000001C0 96 6B 56 62 49 00 00 96 AE C4 23 40 80 33 03 F5 ..kVbI.....#@.3..
000001D0 9A 71 0F BF C8 80 D2 9A 71 13 CF C1 01 E0 4C DE ..q.....q.....L.
000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
000001F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
00000200 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000210 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000220 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000230 3A 09 59 4C 85 87 78 77 88 00 62 38 7B 8D 25 0D ..YL...xw...b8{.%.
00000240 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000250 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000260 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000270 F2 38 D7 8F F4 8F 78 77 88 00 9D C2 82 D4 62 17 ..8....xw.....b.
```

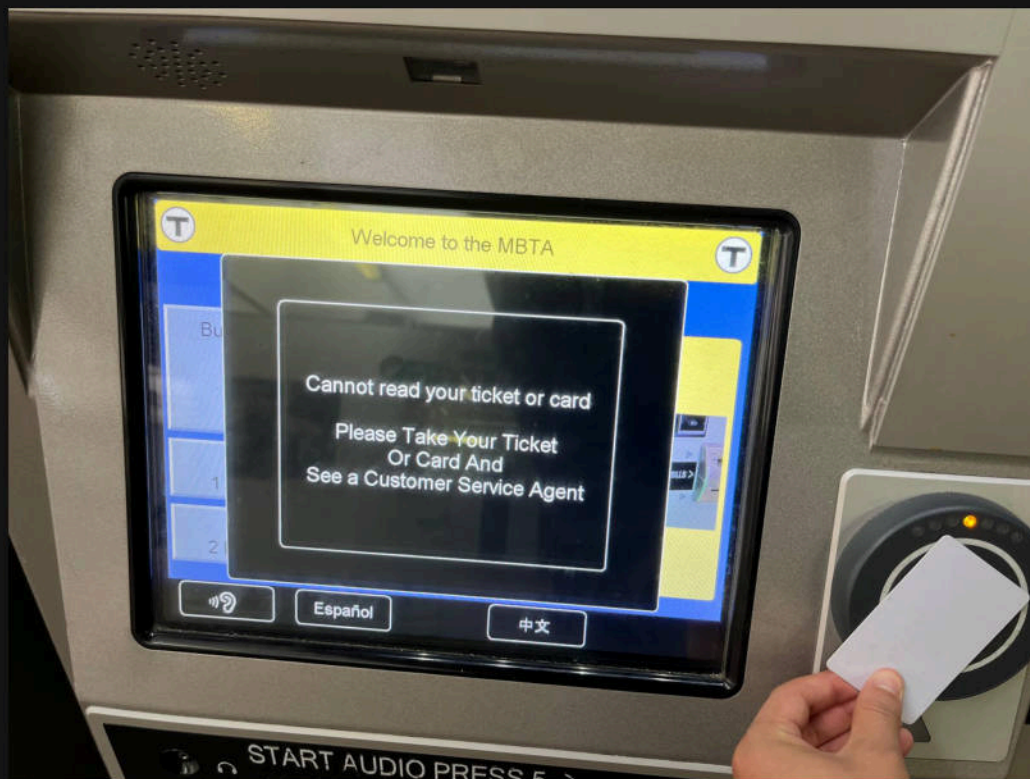
Let's try changing some data

00	20	00	00	00	00	00	00	20	00	00	00	00	00	8E	3F
00	00	00	00	00	00	00	00	00	00	00	00	00	00	86	1F
00	00	00	00	00	00	00	00	00	05	00	00	00	00	82	4B



00	20	00	00	00	00	00	00	20	00	00	00	00	00	8E	3F
13	37	DE	AD	BE	EF	00	00	00	00	00	00	00	00	86	1F
00	00	00	00	00	00	00	00	00	05	00	00	00	00	82	4B

hrm



Smells like a checksum

```
00000000 04 48 5A 35 23 88 04 00 C8 07 00 20 00 00 00 20 HZ5#.....
00000010 4E 0F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030 30 60 20 6F 5B 0A 78 77 88 C1 F1 B9 F5 66 9C C8 0 o[.xw....f..
00000040 04 10 23 45 66 77 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000050 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00
00000060 00 20 20 00 00 00 00 00 00 00 00 00 00 00 00
00000070 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000080 11 9A 71 13 CF C0 94 B7 E9 65 00 0F 08 00 F0 0D ..q.....e.....
00000090 5B FC 9E CD 00 00 FA 00 19 F3 4E 31 4C 78 52 E9 [.....N1LxR.
000000A0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .....>[
000000B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
000000C0 11 9A 71 13 CF C0 94 B7 E9 65 00 0E 80 00 8C FA ..q.....e.....
000000D0 5B FC 9E CD 00 00 C8 00 19 F3 4E 31 4C 78 A0 33 [.....N1Lx.3
000000E0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B .....>[
000000F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000100 00 20 00 00 00 00 00 00 00 00 00 00 00 00 8E 3F .....?
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
00000120 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B .....K
00000130 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000140 00 20 00 00 00 00 00 00 00 00 00 00 00 00 8E 3F .....?
00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
00000160 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B .....K
00000170 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
00000180 9A 82 B4 BF D8 80 C8 9A 82 B5 BF D8 80 32 D8 54 .....2.T
00000190 96 1C A0 60 B9 01 54 96 1C A8 60 B9 01 54 68 3F .....T...Th?
000001A0 96 65 D8 61 21 01 54 96 6B 49 13 D1 01 54 49 D6 ..e.a!.T.kI...TI.
000001B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
000001C0 96 6B 56 62 49 00 00 96 AE C4 23 40 80 33 03 F5 ..kVbI.....#@.3..
000001D0 9A 71 0F BF C8 80 D2 9A 71 13 CF C1 01 E0 4C DE ..q.....q....L.
000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F .....
000001F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$...
00000200 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000210 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000220 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000230 3A 09 59 4C 85 87 78 77 88 00 62 38 7B 8D 25 0D ..YL...xw...b8{.%.
00000240 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000250 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000260 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
00000270 F2 38 D7 8F F4 8F 78 77 88 00 9D C2 82 D4 62 17 ..8....xw.....b.
```


A checksum? Can't be too hard to crack, right?

- Maybe it's a standard checksum
- Why wouldn't they use a standard, secure algorithm?
- Let's try some common algorithms!



Some common algorithms

metool

Home Coding Tool Encrypt/Decrypt Other Tool Search

Coding Tool > Longitudinal Redundancy Check (LRC) Calculator

LRC check calculator, also known as longitudinal redundancy check calculator, is a common error code check tool in the communication field, which can be used for LRC algorithm check of Modbus protocol ASCII mode.

Popular Tools:

- Online Checksum Calculator
- XOR check_BCC check calculator
- Adler32 checksum calculator

Recommend Tool:

- Adler32 checksum calculator
- XOR check_BCC check calculator
- Online Checksum Calculator

CRC Calculator BCC Calculator LRC Calculator CheckSum Calculator Adler32 Calculator

Hex Ascii

Data to be verified: 00 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00

The input data is hexadecimal, for example: 36 37 61 46

LRC Calculate 14 Bytes

Calculation result (Hex): 00

Calculation result (Dec): 192

Calculation result (Oct): 300

Calculation result (Bin): 1100 0000

GIGAcalculator

Calculators Converters Randomizers Articles Find calculator/convert

CRC Calculator

Use this calculator to easily calculate the CRC-32, CRC-16 or CRC-8 hash of a given string. Check CRC using this CRC generator.

CRC algorithm: CRC-8 CRC-16 CRC-32

String: 00200000000000000000000000000000

Calculate

Calculation results

CRC-16 hash: 0x00FF

Share calculator: get direct link

Embed this tool: get code

Cryptography calculators

- md5 Online Generator
- SHA1 Online Generator
- SHA256 Online Generator
- SHA512 Online Generator
- CRC Calculator

Online Checksum Calculator

This Checksum Calculator allows you to find the checksum of your input string. The entered ASCII or Hex string will produce a checksum value that can be used to verify the checksum algorithm used by a particular device. This tool is especially useful for interfacing with devices for IoT and sensor-to-cloud applications.

Hex Input: #20000000000000000000000000000000

ASCII Input: *****

Checksum 8-bit

Checksum 8-bit Normal: 00

Checksum Modulo 256

Sum of Bytes % 256 Normal: 40

Checksum 2s Complement

2s100 - Sum of Bytes Normal: 00

CRC-16 (MODBUS)

Generator Type	Big Endian (MSB) Little Endian (LSB)
Normal	00 01 01 00
Reversed	-- --
Reversed Reversed	-- --

CRC-16-CCITT (X.25, V.41, HDLC, XMODEM, Bluetooth, SD, many others. Known as CRC-CCITT)

Generator Type	Big Endian (MSB) Little Endian (LSB)
Normal (as 1601)	F7 07 07 F7
Reversed (as 1602)	C6 36 36 C6
Reversed Reversed (as 1603)	34 76 76 34

md5Calc.com

Home About Contact Login

Hash calculator

Home Hash calculator MD5 hash for "00200000000000000000000000000000"

IP and Browser >

Internet >

Text >

Minimizers >

Obfuscators >

Random >

Math >

BASE64 Encode/Decode >

JSON Encode/Decode >

URL Encode/Decode >

MD5 hash is

"d1e919e7384c5f3e806524138cf01f4e"

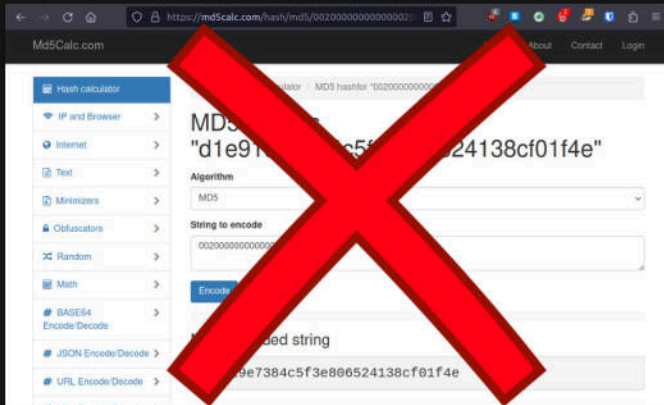
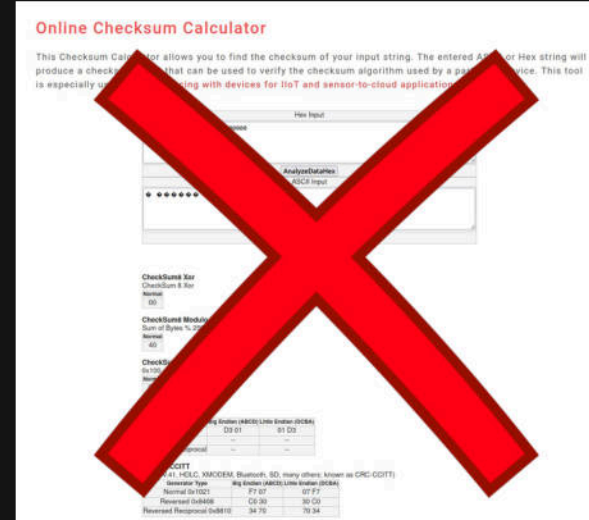
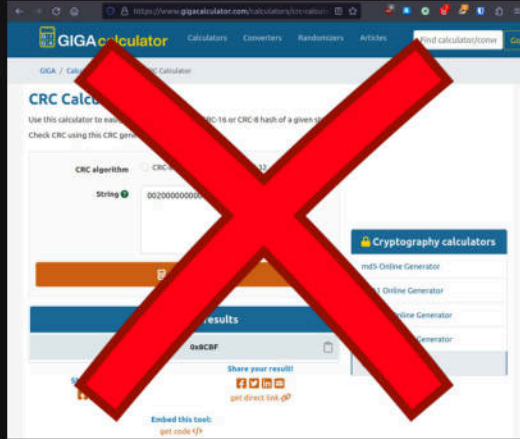
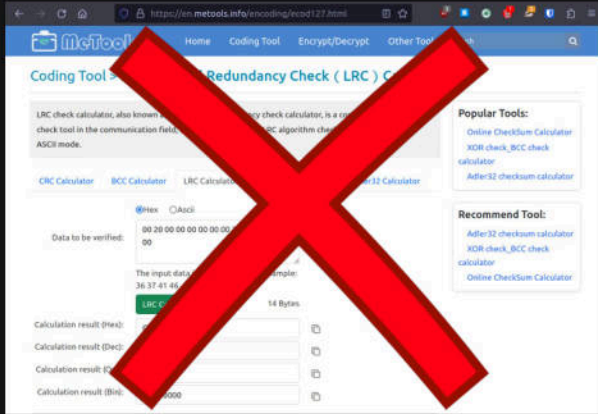
Algorithm: MD5

String to encode: 00200000000000000000000000000000

Encode

MD5 encoded string: d1e919e7384c5f3e806524138cf01f4e

Some common algorithms



What now?

- This checksum is evil
- Maybe cloning cards ain't so bad
- Maybe I'll just put the project to rest



November 2021



Then One Day...



February 2022

Let's figure out where the money is!

Let's stare at some hex data together!

00000010	4E 0F 04 10	04 10 04 10	04 10 04 10	04 10 04 10	N.....
00000020	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000030	30 60 20 6F	5B 0A 78 77	88 C1 F1 B9	F5 66 9C C8	0` o[.xw.....f..
00000040	04 10 23 45	66 77 00 00	00 00 00 00	00 00 00 00	..#Efw.....
00000050	00 03 80 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000060	00 02 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000070	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
00000080	11 89 1C 00	02 90 89 1C	00 65 00 00	00 00 85 9C	e.....
00000090	5B FB 2C 03	E0 00 00 00	00 20 00 00	00 00 15 2D	[.,.....-.
000000A0	00 20 00 00	00 00 00 00	00 00 00 00	00 00 28 E3	(.
000000B0	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
000000C0	11 9A 32 AA	BF F8 9A 32	AA 65 00 00	90 00 1A E3	..2....2.e.....
000000D0	5B FD 4E D9	20 00 32 00	00 20 00 00	00 00 BD 1E	[.N. .2..
000000E0	00 20 00 00	00 00 00 00	00 00 00 00	00 00 28 E3	(.
000000F0	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
00000100	00 20 00 00	00 00 00 00	20 00 00 00	00 00 88 C3	

For hours

00000010	4E 0F 04 10	04 10 04 10	04 10 04 10	04 10 04 10	N.....
00000020	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000030	30 60 20 6F	5B 0A 78 77	88 C1 F1 B9	F5 66 9C C8	0` o[.xw.....f..
00000040	04 10 23 45	66 77 00 00	00 00 00 00	00 00 00 00	..#Efw.....
00000050	00 03 80 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000060	00 02 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000070	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
00000080	11 89 1C 00	02 90 89 1C	00 65 00 00	00 00 85 9C	e.....
00000090	5B FB 2C 03	E0 00 00 00	00 20 00 00	00 00 15 2D	[.,.....-.
000000A0	00 20 00 00	00 00 00 00	00 00 00 00	00 00 28 E3	(.
000000B0	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
000000C0	11 9A 32 AA	BF F8 9A 32	AA 65 00 00	90 00 1A E3	..2....2.e.....
000000D0	5B FD 4E D9	20 00 32 00	00 20 00 00	00 00 BD 1E	[.N. .2..
000000E0	00 20 00 00	00 00 00 00	00 00 00 00	00 00 28 E3	(.
000000F0	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
00000100	00 20 00 00	00 00 00 00	20 00 00 00	00 00 88 C3	



and hours

00000010	4E 0F 04 10	04 10 04 10	04 10 04 10	04 10 04 10	N.....
00000020	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000030	30 60 20 6F	5B 0A 78 77	88 C1 F1 B9	F5 66 9C C8	0` o[.xw.....f..
00000040	04 10 23 45	66 77 00 00	00 00 00 00	00 00 00 00	..#Efw.....
00000050	00 03 80 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000060	00 02 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
00000070	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
00000080	11 89 1C 00	02 90 89 1C	00 65 00 00	00 00 85 9C	e.....
00000090	5B FB 2C 03	E0 00 00 00	00 20 00 00	00 00 15 2D	[.,.....-.
000000A0	00 20 00 00	00 00 00 00	00 00 00 00	00 00 28 E3	(.
000000B0	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
000000C0	11 9A 32 AA	BF F8 9A 32	AA 65 00 00	90 00 1A E3	..2....2.e.....
000000D0	5B FD 4E D9	20 00 32 00	00 20 00 00	00 00 BD 1E	[.N. .2..
000000E0	00 20 00 00	00 00 00 00	00 00 00 00	00 00 28 E3	(.
000000F0	5E C3 9B 02	2F 2B 78 77	88 00 F6 62	24 8E 7E 89	^.../+xw...b\$.~.
00000100	00 20 00 00	00 00 00 00	20 00 00 00	00 00 88 C3	





**2000 YEARS
LATER**

We Found Nothing

How do you reverse engineer things again?

reverse engineering

**Everybody talks about it,
But where do you start?**

- 1) Make a guess about what's in the data
- 2) Change a single variable; see what changes
- 3) Repeat many times with varying data
- 4) Compare similar and dissimilar data
- 5) Ignore constant regions
- 6) Build/use tools

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Maybe Let's Try Another Crack at that Checksum

00	20	00	00	00	00	00	00	20	00	00	00	00	00	00	88	C3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	05	00	00	00	00	00	84	B7
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89	

00	20	00	00	00	00	00	00	20	00	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	

Maybe It's an Off The Shelf Checksum... With Salt

metool

Home Coding Tool Encrypt/Decrypt Other Tool Search

Coding Tool > Longitudinal Redundancy Check (LRC) Calculator

LRC check calculator, also known as longitudinal redundancy check calculator, is a common error code check tool in the communication field, which can be used for LRC algorithm check of Modbus protocol ASCII mode.

Popular Tools:

- Online Checksum Calculator
- XOR check_BCC check calculator
- Adler32 checksum calculator

Recommend Tool:

- Adler32 checksum calculator
- XOR check_BCC check calculator
- Online Checksum Calculator

CRC Calculator BCC Calculator LRC Calculator CheckSum Calculator Adler32 Calculator

Hex Ascii

Data to be verified:

00 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00

The input data is hexadecimal, for example: 36 37 41 46

LRC Calculate 14 Bytes

Calculation result (Hex): 00

Calculation result (Dec): 192

Calculation result (Oct): 300

Calculation result (Bin): 1100 0000

GIGAcalculator

Calculators Converters Randomizers Articles Find calculator/convert

CRC Calculator

Use this calculator to easily calculate the CRC-32, CRC-16 or CRC-8 hash of a given string. Check CRC using this CRC generator.

CRC algorithm: CRC-8 CRC-16 CRC-32

String: 00200000000000000000000000000000

Calculate

Calculation results

CRC-16 hash: 0x3CBF

Share calculator: get direct link

Embed this tool: get code

Cryptography calculators

- md5 Online Generator
- SHA1 Online Generator
- SHA256 Online Generator
- SHA512 Online Generator
- CRC Calculator

Online Checksum Calculator

This Checksum Calculator allows you to find the checksum of your input string. The entered ASCII or Hex string will produce a checksum value that can be used to verify the checksum algorithm used by a particular device. This tool is especially useful for interfacing with devices for **IIoT** and **sensor-to-cloud** applications.

Hex Input

ASCII Input

Checksum 8-bit

Checksum Modulo 256

Checksum 2's Complement

CRC-16 (MODBUS)

CRC-16 CCITT

Generator Type	Big Endian (MSB/LSB) Input (HEX)	Big Endian (MSB/LSB) Output (HEX)
Normal	00 01	01 00
Reversed	00 01	00 01
Reversed (Reversed)	00 01	00 01

Generator Type	Big Endian (MSB/LSB) Input (HEX)	Big Endian (MSB/LSB) Output (HEX)
Normal (CCITT)	F7 07	07 F7
Reversed (CCITT)	C6 30	30 C6
Reversed (Reversed (CCITT))	34 76	76 34

md5Calc.com

Home About Contact Login

Hash calculator

MD5 hash is "d1e919e7384c5f3e806524138cf01f4e"

Algorithm: MD5

String to encode: 00200000000000000000000000000000

Encode

MD5 encoded string: d1e919e7384c5f3e806524138cf01f4e

IP and Browser

Internet

Text

Minimizers

Obfuscators

Random

Math

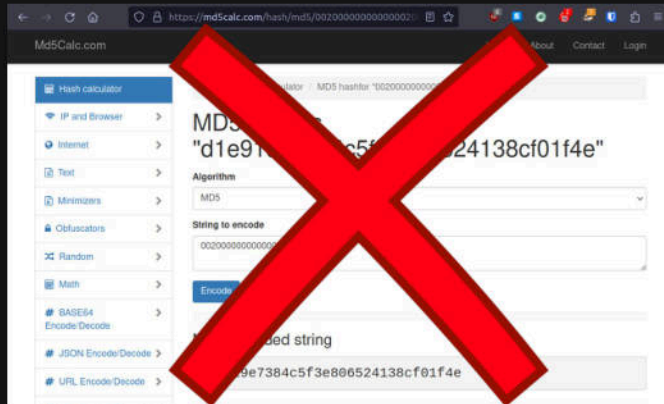
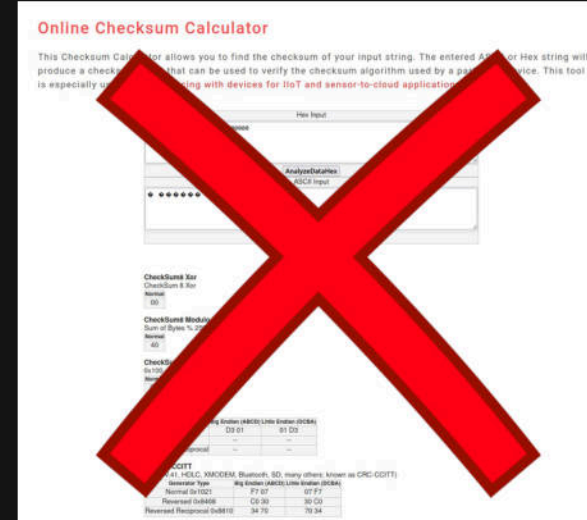
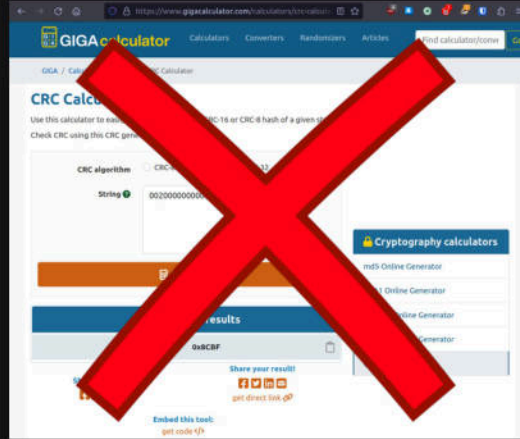
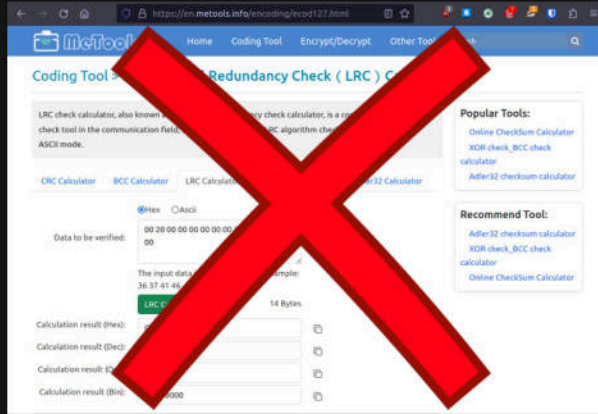
BASE64 Encode/Decode

JSON Encode/Decode

URL Encode/Decode



Or not



~~English Class~~ CharlieCard Hacking Class

- What to do during English class
- Learn... No
- Work... No
- Stare at hex data and try random checksum algorithms... Yes!



Our First Breakthrough

March 2022

00	20	00	00	00	00	00	00	20	00	00	00	00	00	00	88	C3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	05	00	00	00	00	00	84	B7
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89	

00	20	00	00	00	00	00	00	20	00	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	

Card 1

00 00 00 00 00 00 00 00 00 00 00 00 00 00 80 E3

XOR = 0x98C3

00 00 00 00 00 00 00 00 00 00 00 00 00 00 18 20

Card 2

Card 1

00 00 00 00 00 00 00 00 00 00 00 00 00 00 80 E3

XOR = 0x98C3

Card 2

00 00 00 00 00 00 00 00 00 00 00 00 00 00 18 20

Card 1

00 20 00 00 00 00 00 00 20 00 00 00 00 00 88 C3

XOR = 0x98C3

Card 2

00 20 00 00 00 00 00 00 20 00 00 00 00 00 10 00

So How Do We Use This?

Find Two Identical Lines

5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	05	00	00	9A	32	AA	BF	FD	00	32	9A	53
9A	32	AA	BF	FD	00	32	00	00	00	00	00	00	00	FF	11
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF
FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF

5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	05	00	00	9E	37	AF	C0	8D	00	32	92	23
9E	37	AF	C0	8D	00	32	00	00	00	00	00	00	00	7E	9A
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF
FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF

- Find two lines with the same data on two different cards
- Note that identical lines on different cards have a different checksum



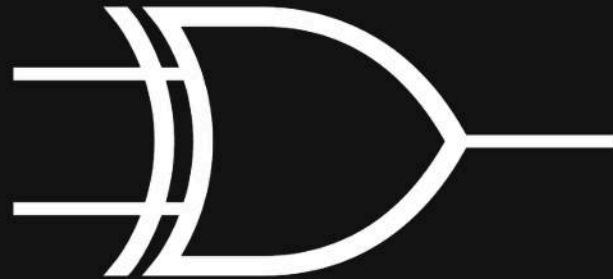
Find Checksum Modifier

5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	05	00	00	9A	32	AA	BF	FD	00	32	9A	53
9A	32	AA	BF	FD	00	32	00	00	00	00	00	00	00	FF	11
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF
FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF

- XOR the checksums
- The value you get is the “checksum modifier” between the two cards

80E3
XOR 1820
98C3

5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	05	00	00	9E	37	AF	C0	8D	00	32	92	23
9E	37	AF	C0	8D	00	32	00	00	00	00	00	00	00	7E	9A
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20



Copy Any Line From Old Card to New Card

5B	FC	6B	3B	E0	00	00	00	00	20	00	00	00	00	7E	7E
00	20	00	00	00	00	00	00	00	00	00	00	00	00	B0	20
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
11	9E	37	AF	C0	88	9E	37	AF	65	00	00	90	00	0E	DB
5B	FD	CF	79	C0	00	32	00	00	20	00	00	00	00	8A	85
00	20	00	00	00	00	00	00	00	00	00	00	00	00	B0	20
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	10	00
00	00	00	00	00	00	00	00	00	00	00	00	00	00	18	20
00	00	00	00	00	00	00	00	00	05	00	00	00	00	1C	74
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89

Card 1



5B	FB	2C	03	E0	00	00	00	00	20	00	00	00	00	15	2D
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
11	9A	32	AA	BF	F8	9A	32	AA	65	00	00	90	00	1A	E3
5B	FD	CF	79	C0	00	32	00	00	20	00	00	00	00	8A	85
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89

Card 2

XOR Checksum With Checksum Modifier To Get New Checksum

$$0x8A85 \text{ XOR } 0x98C3 = 0x1246$$

5B	FB	2C	03	E0	00	00	00	00	20	00	00	00	00	00	15	2D
00	20	00	00	00	00	00	00	00	00	00	00	00	00	00	28	E3
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
11	9A	32	AA	BF	F8	9A	32	AA	65	00	00	90	00	1A	E3	
5B	FD	CF	79	C0	00	32	00	00	20	00	00	00	00	8A	85	
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3	
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3	
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3	
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3	
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89	
00	00	00	00	05	00	00	9A	32	AA	BF	FD	00	32	9A	53	



5B	FB	2C	03	E0	00	00	00	00	20	00	00	00	00	00	15	2D
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
11	9A	32	AA	BF	F8	9A	32	AA	65	00	00	90	00	1A	E3	
5B	FD	CF	79	C0	00	32	00	00	20	00	00	00	00	12	46	
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3	
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3	
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3	
00	00	00	00	00	00	00	00	00	00	00	00	00	00	80	E3	
00	00	00	00	00	00	00	00	00	05	00	00	00	00	84	B7	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	EE	1E	89	

BOOM! LINE = COPIED

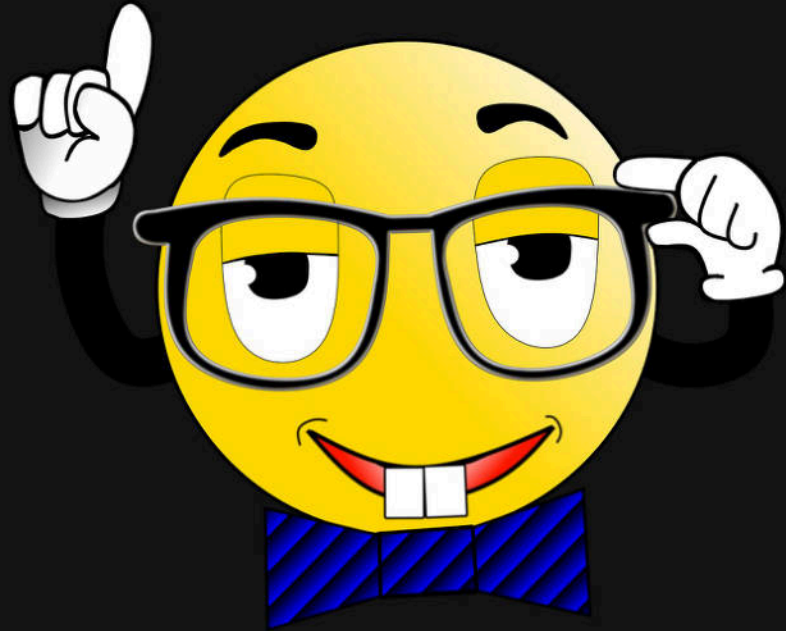


Instructions

- **Step 1: Start with two cards**
- **Step 2: Find two identical lines**
- **Step 3: XOR the two different checksums to get checksum modifier**
- **Step 4: Find a line you want to move**
- **Step 5: Copy that line onto the other card**
- **Step 6: XOR the old checksum by checksum modifier**
- **Step 7: Profit**

What can we do now?

- Copy money values from card to card
- Copy anything from card to card
- Better than cloning, can't get disabled



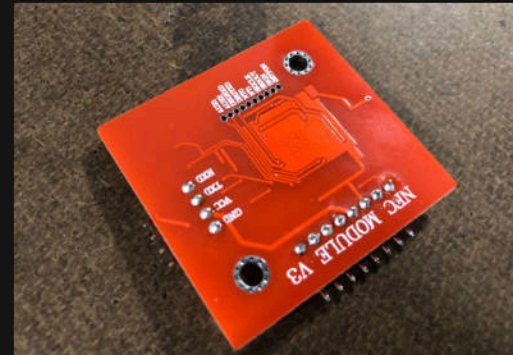
Next step: Trial and error

Subway Station Adventure Time!

March 2022

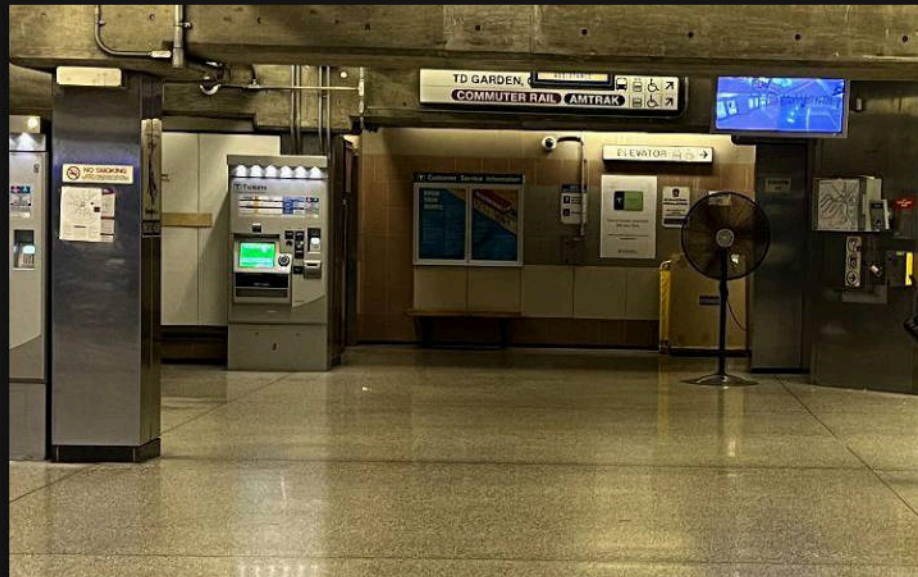
Step 1: Make an NFC Reader Contraption

- Ingredients: PN532, Raspberry Pi, Battery Bank
- Step 1: Hook up PN532 to Raspberry Pi
- Step 2: Plug battery bank into Raspberry Pi
- Step 3: Make your own WiFi hotspot and SSH into the Raspberry Pi
- Step 4: Toss contraption in backpack





Step 2: Spend hours days at a train station



Both Stations Smell Funny



Step 3: Mess around

- Make ~~random~~ educated guess-based changes
- Flash changes onto cards
- Tap cards
- Enjoy error noise
- Try to not make it obvious you're up to no good

Another Breakthrough

March 2022

Take A Look At These Bytes

Current Value:
\$1.25

Last Value:
\$1.00

Current Value:
\$2.15

Last Value:
\$1.35

77-30-125.mrfd																
0000	0010	4E	9F	84	10	84	10	84	10	84	10	84	10	84	10	N.....
0000	0020	80	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000	0030	30	60	20	6F	58	8A	78	77	88	C1	F1	B9	F5	66	9C CB
0000	0040	84	10	23	45	66	77	88	00	00	00	00	00	00	00	00
0000	0050	80	1F	A0	00	00	00	00	00	00	00	00	00	00	00	00
0000	0060	80	20	20	00	00	00	00	00	00	00	00	00	00	00	00
0000	0070	5E	C3	9B	92	2F	28	78	77	88	00	F6	62	24	8E	7E 89
0000	0080	11	9A	F1	13	8F	10	84	07	E9	65	00	00	00	00	7F 00
0000	0090	58	FC	9E	CD	00	00	00	00	13	F1	4E	31	4C	78	32 1E
0000	00A0	60	20	00	00	00	00	00	00	00	00	04	00	00	00	3E 58
0000	00B0	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	8E	7E 89
0000	00C0	11	9A	F1	13	8F	10	84	07	E9	65	00	00	00	00	8C 3A
0000	00D0	58	FC	9E	CD	00	00	00	00	13	F1	4E	31	4C	78	40 12
0000	00E0	60	20	00	00	00	00	00	00	00	00	04	00	00	00	3E 58
0000	00F0	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	8E	7E 89
0000	0100	60	20	00	00	00	00	00	00	28	00	00	00	00	00	8E 3F
0000	0110	80	00	00	00	00	00	00	00	88	00	00	00	00	00	00
0000	0120	80	00	00	00	00	00	00	00	00	65	00	00	00	00	82 4B
0000	0130	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	8E	7E 89
0000	0140	60	20	00	00	00	00	00	00	28	00	00	00	00	00	8E 3F
0000	0150	80	00	00	00	00	00	00	00	00	65	00	00	00	00	82 4B
0000	0160	80	00	00	00	00	00	00	00	00	65	00	00	00	00	82 4B
0000	0170	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	EE	1E 89
07-30-215.mrfd																
0000	0010	4E	9F	84	10	84	10	84	10	84	10	84	10	84	10	N.....
0000	0020	80	00	00	00	00	00	00	00	00	00	00	00	00	00	
0000	0030	30	60	20	6F	58	8A	78	77	88	C1	F1	B9	F5	66	9C CB
0000	0040	84	10	23	45	66	77	88	00	00	00	00	00	00	00	00
0000	0050	80	1F	A0	00	00	00	00	00	00	00	00	00	00	00	00
0000	0060	80	1C	20	00	00	00	00	00	00	00	00	00	00	00	00
0000	0070	5E	C3	9B	92	2F	28	78	77	88	00	F6	62	24	9E	7E 89
0000	0080	11	95	88	38	83	40	84	07	E9	65	00	00	00	00	8E 8E
0000	0090	58	FC	9E	CD	00	00	00	00	00	00	00	00	00	00	4F 78
0000	00A0	60	20	00	00	00	00	00	00	00	00	00	00	00	00	2E 1F
0000	00B0	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	9E	7E 89
0000	00C0	11	95	88	38	83	40	84	07	E9	65	00	00	00	00	C7 3A
0000	00D0	58	FC	9E	CD	00	00	00	00	00	00	20	00	00	00	0F 78
0000	00E0	60	20	00	00	00	00	00	00	00	00	00	00	00	00	2E 1F
0000	00F0	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	8E	7E 89
0000	0100	60	20	00	00	00	00	00	00	28	00	00	00	00	00	8E 3F
0000	0110	80	00	00	00	00	00	00	00	88	00	00	00	00	00	00
0000	0120	80	00	00	00	00	00	00	00	00	65	00	00	00	00	82 4B
0000	0130	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	8E	7E 89
0000	0140	60	20	00	00	00	00	00	00	28	00	00	00	00	00	8E 3F
0000	0150	80	00	00	00	00	00	00	00	00	65	00	00	00	00	82 4B
0000	0160	80	00	00	00	00	00	00	00	00	65	00	00	00	00	82 4B
0000	0170	5E	C3	90	92	2F	28	78	77	88	00	F6	62	24	EE	1E 89
Arrow keys move F find RET next difference ESC quit T move top C ASCII E edit file G goto position Q quit B move bottom																

\$1.00

\$1.35

0xC8 -> 200

Interesting...

Current Value:
\$1.25

Last Value:
\$1.00

Current Value:
\$2.15

Last Value:
\$1.35

```
07-30-125.mfd
0000 0010: 4E 8F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
0000 0020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0000 0030: 30 60 20 6F 50 8A 78 77 88 C1 F1 89 F5 66 9C C8 0 of_xw f
0000 0040: 04 10 23 45 66 77 80 00 00 00 00 00 00 00 00 00 ..SEW...f
0000 0050: 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0000 0060: 00 20 20 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0000 0070: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 0080: 11 0A F1 13 CF 10 04 67 E9 65 00 00 00 00 00 00 ~...v...v...
0000 0090: 50 FC 9E CD 00 00 00 19 F3 4E 31 4C 78 A0 33 ~...NILLAR
0000 00A0: 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 50 ~...x
0000 00B0: 5E C3 90 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 00C0: 11 0A F1 13 CF 10 04 67 E9 65 00 00 00 00 00 00 ~...v...v...
0000 00D0: 50 FC 9E CD 00 00 00 19 F3 4E 31 4C 78 A0 33 ~...NILLAR
0000 00E0: 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 50 ~...x
0000 00F0: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 0100: 00 20 00 00 00 00 00 00 00 00 20 00 00 00 00 00 .....
0000 0110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1F .....
0000 0120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 4B .....K
0000 0130: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 0140: 00 20 00 00 00 00 00 00 00 00 20 00 00 00 00 00 .....
0000 0150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1F .....
0000 0160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 4B .....K
0000 0170: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ~.../xw...b$~

07-30-215.mfd
0000 0010: 4E 8F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
0000 0020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0000 0030: 30 60 20 6F 50 8A 78 77 88 C1 F1 89 F5 66 9C C8 0 of_xw f
0000 0040: 04 10 23 45 66 77 80 00 00 00 00 00 00 00 00 00 ..SEW...f
0000 0050: 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0000 0060: 00 1E 20 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0000 0070: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 0080: 11 0A 00 04 02 00 04 67 E9 65 00 00 00 00 00 00 ~...v...v...
0000 0090: 50 FC 9E CD 00 00 00 00 20 00 00 00 00 00 00 00 ~...v...v...
0000 00A0: 00 20 00 00 00 00 00 00 00 00 00 00 00 00 2E 1F .....
0000 00B0: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 00C0: 11 0A 00 04 02 00 04 67 E9 65 00 00 00 00 00 00 ~...v...v...
0000 00D0: 50 FC 9E CD 00 00 00 00 00 20 00 00 00 00 00 00 ~...v...v...
0000 00E0: 00 20 00 00 00 00 00 00 00 00 00 00 00 00 2E 1F .....
0000 00F0: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 0100: 00 20 00 00 00 00 00 00 00 00 20 00 00 00 00 00 .....
0000 0110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1F .....
0000 0120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 4B .....K
0000 0130: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ~.../xw...b$~
0000 0140: 00 20 00 00 00 00 00 00 00 00 20 00 00 00 00 00 .....
0000 0150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1F .....
0000 0160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 4B .....K
0000 0170: 5E C3 98 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ~.../xw...b$~

Arrow keys move F find RET next difference ESC quit T move top
C ASCII EBCDIC E edit file G goto position Q quit B move bottom
```

0xFA -> 250 / 2 -> 125

0xC8 -> 200 / 2 -> 100

0x010E -> 270 / 2 -> 135

0x01AE -> 430 / 2 -> 215

Our Findings

- Where the money's stored!!!
- Money stored in half-pennies
- Two transaction registers (current and last money value)
- CharlieCards are weird



Let's Try to Change It

April 2022

Step 1: Isolate variables

0xFA = 250 half pennies = \$1.25

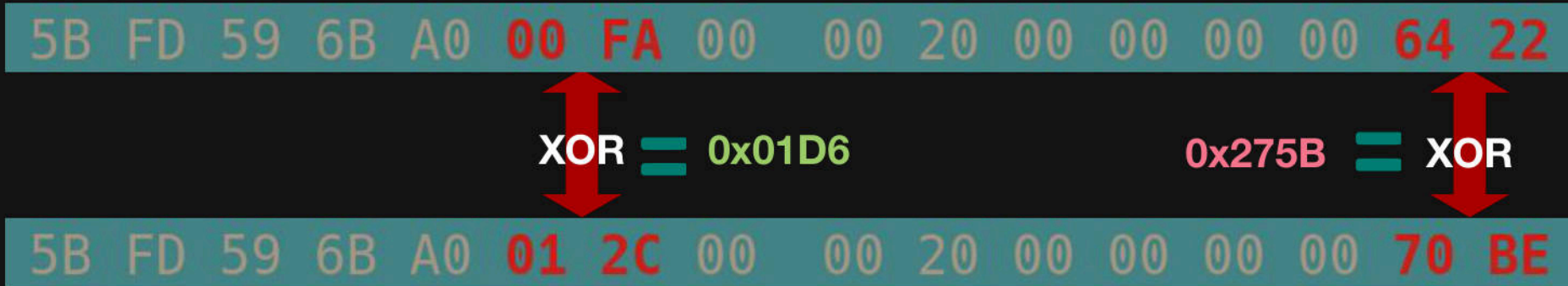
5B FD 59 6B A0 **00 FA** 00 00 20 00 00 00 00 **64 22**

5B FD 59 6B A0 **01 2C** 00 00 20 00 00 00 00 **70 BE**

0x012C = 300 half pennies = \$1.50

Step 2: XOR everything

0xFA = 250 half pennies = \$1.25



0x012C = 300 half pennies = \$1.50

Data Modifier

Checksum Modifier

Step 3: Try this strategy with different money values

$0x0 = 0$ half pennies = \$0.00

5B FD 59 6B A0 00 00 00 00 20 00 00 00 00 00 BA A3

XOR by 0x01D6

XOR by 0x149C

5B FD 59 6B A0 01 D6 00 00 20 00 00 00 00 00 AE 3F

$0x01D6 = 470$ half pennies = \$2.35

CharlieCard

Transit Value - Adult

Remaining Amount: \$ 2.35

Please Make Your Selection

Buy Transit Value

1 or 7 Day LinkPass

Calendar Based
Monthly Passes

Quick Ticket

Card / Ticket
Information



Go
Back

Cancel

START AUDIO PRESS 5 >

Step 4: Add a quarter



Why add a quarter?

$$\begin{array}{ccc} 0b0000_1000 & \text{XOR} & 0b0000_0111 \\ (8) & & (7) \end{array} = 0b0000_1111 \quad (15)$$

$$\begin{array}{ccc} 0b0000_1111 & + & 1 \\ (15) & & \end{array} = 0b0001_0000 \quad (16)$$

$$\begin{array}{ccc} 0b0001_0000 & \text{XOR} & 0b0000_1111 \\ (16) & & (15) \end{array} = 0001_1111 \quad (31)$$

Step 5: Inch your way up

0x01D6 = 470 half pennies = \$2.35

5B FD 59 6B A0 **01 D6** 00 00 20 00 00 00 00 00 **AE 3F**

XOR = 0x03DE

0x94B6 = **XOR**

5B FD 59 6B A0 **02 08** 00 00 20 00 00 00 00 00 **3A 89**

0x0208 = 520 half pennies = \$2.60

Step 5: Inch your way up

0x00 = 0 half pennies = \$0.00

5B FD 59 6B A0 00 00 00 00 20 00 00 00 00 BA A3

XOR by 0x03DE

XOR by 0x94B6

5B FD 59 6B A0 03 DE 00 00 20 00 00 00 00 2E 15

0x03DE = 990 half pennies = \$4.95

CharlieCard

Transit Value - Adult

Remaining Amount: \$ 4.95

Please Make Your Selection

Buy Transit Value

1 or 7 Day LinkPass

Calendar Based
Monthly Passes

Quick Ticket

Card / Ticket
Information



Go
Back

Cancel

START AUDIO PRESS 5 >

TICKETS / CHANGE >



Rinse and Repeat



CharlieCard

Transit Value - Adult

Remaining Amount: \$ 163.84

Please Make Your Selection

Buy Transit Value

1 or 7 Day LinkPass

Calendar Based
Monthly Passes

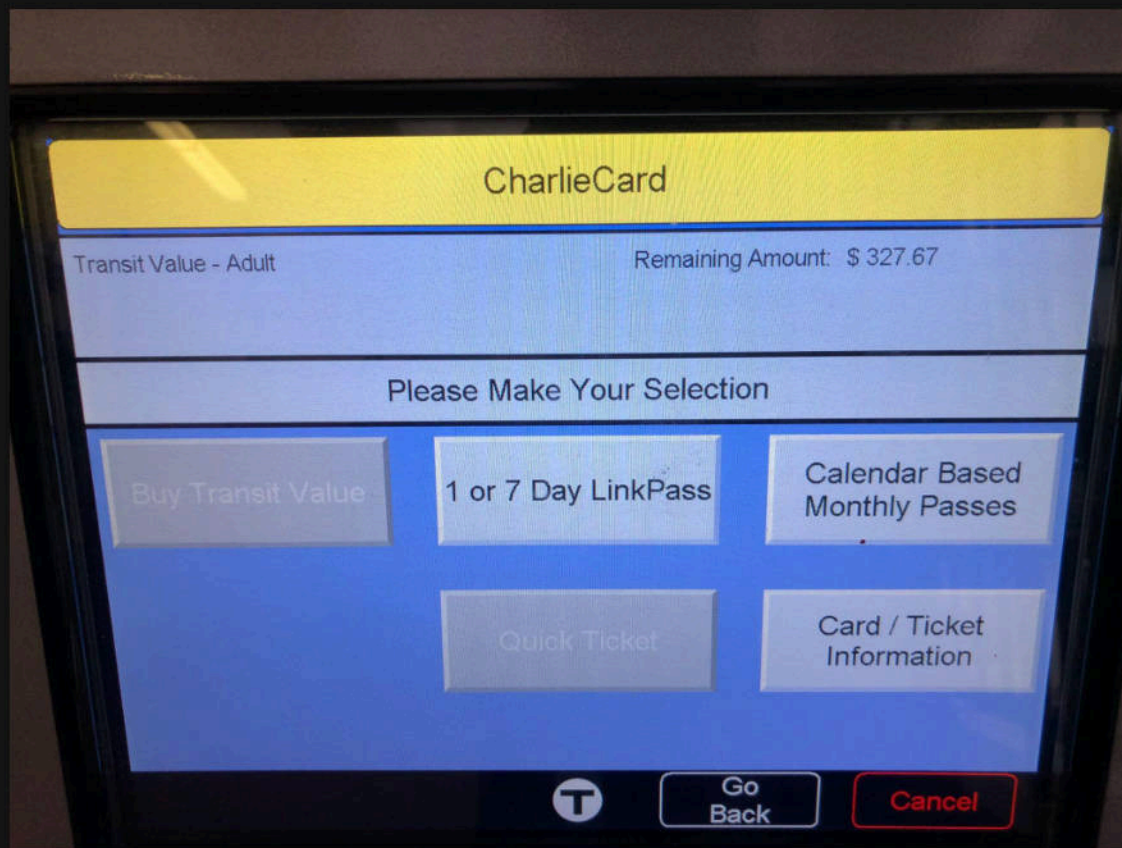
Quick Ticket

Card / Ticket
Information



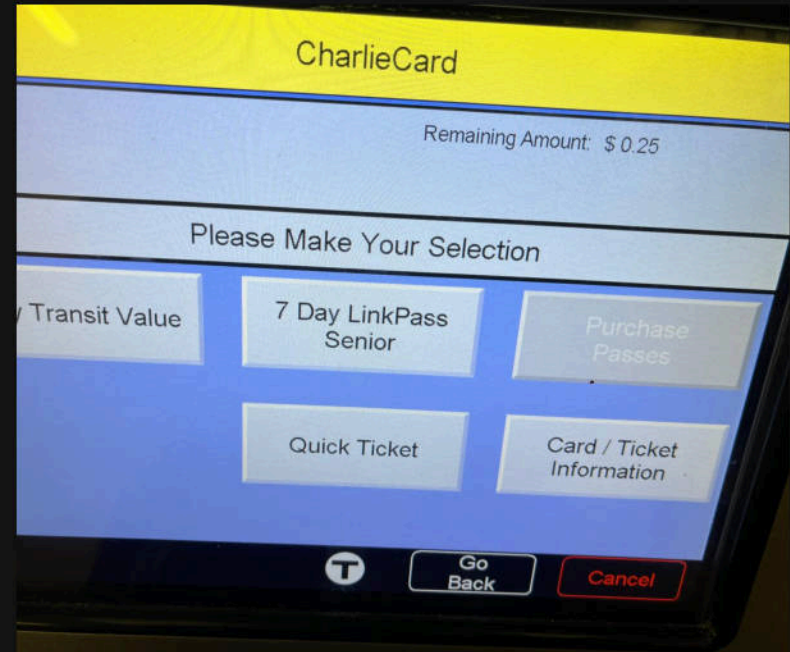
Go
Back

Cancel



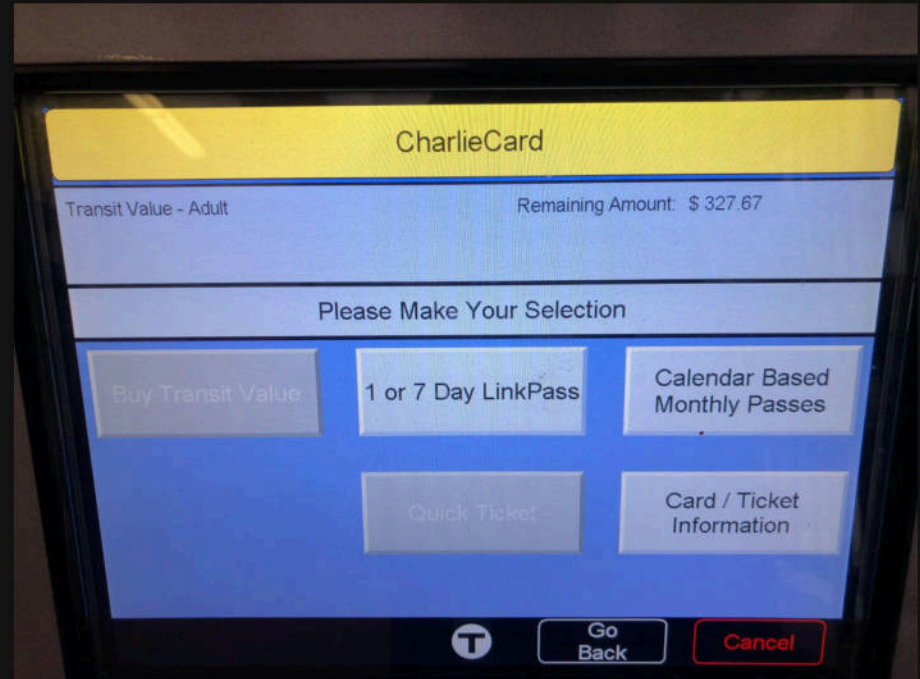
Free rides for life: A free step process

- Step 1: Buy a card for \$0.25



Free rides for life: A free step process

- Step 1: Buy a card for \$0.25
- Step 2: Set the value to \$327.67



Free rides for life: A free step process

- Step 1: Buy a card for \$0.25
- Step 2: Set the value to \$327.67
- Step 3: Profit!



How Does This Work?

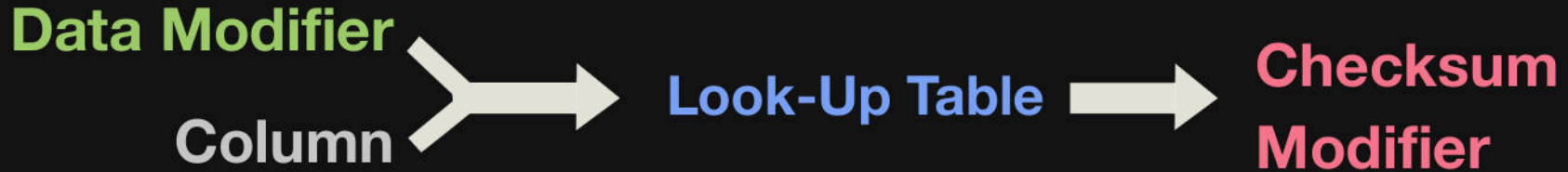
Vocab

1. **Existing Data** -> Pre-existing data that you want to edit
2. **Target Data** -> Value you want the data to be
3. **Data Modifier** -> Value you need to XOR data by to get **Target Data** Value
4. **Existing Checksum** -> You know what this is
5. **Target Checksum** -> Value you need to make the **Target Data** valid
6. **Checksum Modifier** -> Value you need to XOR the checksum with to make it valid
7. **Look-Up Table** -> A super-secret black box that takes in a **Data Modifier** and a column and returns a **Checksum Modifier**

What's a column?

00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
30	60	20	6F	5B	0A	78	77	88	C1	F1	B9	F5	66	9C	C8	
04	10	23	45	66	77	00	00	00	00	00	00	00	00	00	00	
00	03	80	00	00	00	00	00	00	00	00	00	00	00	00	00	
00	02	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
11	89	1C	00	02	90	89	1C	00	65	00	00	00	00	85	9C	
5B	FB	2C	03	E0	00	00	00	00	20	00	00	00	00	15	2D	
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
11	9A	32	AA	BF	F8	9A	32	AA	65	00	00	90	00	1A	E3	
5B	FD	4E	D9	20	00	32	00	00	20	00	00	00	00	BD	1E	
00	20	00	00	00	00	00	00	00	00	00	00	00	00	28	E3	
5E	C3	9B	02	2F	2B	78	77	88	00	F6	62	24	8E	7E	89	
00	20	00	00	00	00	00	00	20	00	00	00	00	00	88	C3	

Existing Data XOR Target Data = Data Modifier



Existing Checksum XOR Checksum Modifier = Target Checksum

Example With Steps

1. Choose What to Change: We Want to Change **0x32 (\$0.25)** to **0xC8 (\$1)** in Column 7
2. Find **Data Modifier**: XOR **0x32** with **0xC8** getting **0xFA**
3. Find **Checksum Modifier**: Put **0xFA** and Column 7 into the **Lookup Table**
4. XOR **Checksum Modifier (0xDE81)** with the **Existing Checksum** to get the **Target Checksum**

A few more station adventures later...

Anatomy of a CharlieCard

UID

Number of Uses

Money

Expiration Date

Card Type

Transaction History

Timestamp

Location

```
00000000 04 48 5A 35 23 88 04 00 C8 07 00 20 00 00 00 20 HZ5#.....
00000010 4E 0F 04 10 04 10 04 10 04 10 04 10 04 10 04 10 N.....
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..o[.xw....f..
00000030 30 60 20 6F 5B 0A 78 77 88 C1 F1 B9 F5 66 9C C8 ..#Efw.....
00000040 04 10 23 45 66 77 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000050 00 1F A0 00 00 00 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000060 00 20 20 00 00 00 00 00 00 00 00 00 00 00 00 00 ..#Efw.....
00000070 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000080 11 9A 71 13 CF C0 94 B7 E9 65 00 0F 08 00 7F 0D ..q.....e.....
00000090 5B FC 9E CD 00 00 FA 00 19 F3 4E 31 4C 78 52 E9 [.....N1LxR.
000000A0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B ..>[
000000B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
000000C0 11 9A 71 13 CF C0 94 B7 E9 65 00 0E 80 00 8C FA ..q.....e.....
000000D0 5B FC 9E CD 00 00 C8 00 19 F3 4E 31 4C 78 AB 33 [.....N1Lx.3
000000E0 00 20 00 00 00 00 00 00 00 00 04 00 00 00 3E 5B ..>[
000000F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000100 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F ..?
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F ..
00000120 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B ..K
00000130 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 8E 7E 89 ^.../+xw...b$.~.
00000140 00 20 00 00 00 00 00 00 20 00 00 00 00 00 8E 3F ..?
00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F ..
00000160 00 00 00 00 00 00 00 00 00 05 00 00 00 00 82 4B ..K
00000170 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$.~.
00000180 00 82 84 BF 08 80 C8 9A 82 65 BF D8 80 32 D8 54 .....2.T
00000190 96 1C A0 60 00 01 54 96 1C A8 60 B9 01 54 68 3F .....T.....TI?
000001A0 96 65 D8 61 21 01 54 96 6B 49 13 D1 01 54 49 D6 ..e.a!..T.kI...TI.
000001B0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$.~.
000001C0 96 6B 56 62 49 00 00 96 AE C4 23 40 80 33 03 F5 ..kVbI.....#@.3..
000001D0 9A 71 0F BF C8 80 D2 9A 71 13 CF C1 01 E0 4C DE ..q.....q.....L.
000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 86 1F ..
000001F0 5E C3 9B 02 2F 2B 78 77 88 00 F6 62 24 EE 1E 89 ^.../+xw...b$.~.
00000200 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF ..
00000210 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF ..
00000220 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF ..
00000230 3A 09 59 4C 85 87 78 77 88 00 62 38 7B 8D 25 0D ..YL...xw...b8(.%.
00000240 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF ..
00000250 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF ..
00000260 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF ..
00000270 F2 38 D7 8F F4 8F 78 77 88 00 9D C2 82 D4 62 17 ..8....xw.....b.
```

Part 4: Fun Gadgets

Then One Day



April 2022

Vending Machine v1

- Based off the readers on busses
- Fully modeled in CAD
- An abomination
- Impossible for us to manufacture



Vending Machine v2

- Still based off of MBTA bus readers
- Fully modeled in CAD *and cardboard*
- Made with manufacturing in mind
- Hard to manufacture
- Couldn't figure out the *slant*



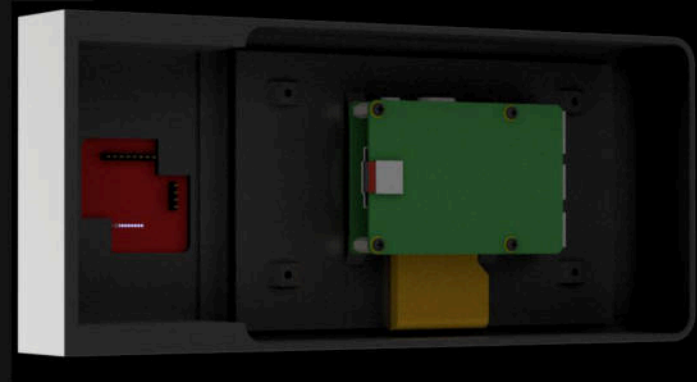
Vending Machine v3

- Still trying to look like bus readers
- Used 3D printed connectors
- Beats the *slant*
- ugly.



Vending Machine v4

- The bus readers are ugly anyway
- Slim
- Larger and more responsive touchscreen
- Giant waste of filament
- Impossible to actually mount anything

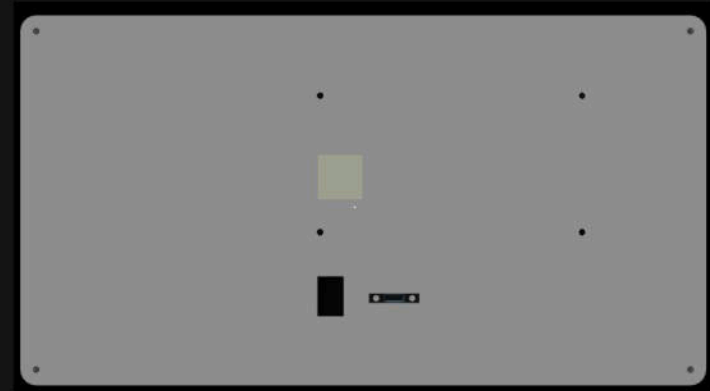
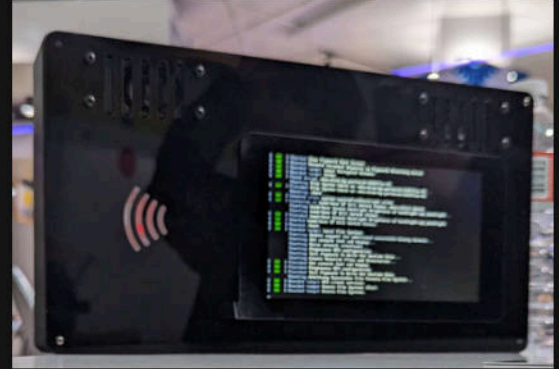


Vending Machine v5 (CharlieKiosk)

- Screen mounted upside down :(
- Actually possible to mount parts???
- Sexy
- Baller speaker system
- Couldn't mount usb-c :(

I broke it

February 2023



Vending Machine v6 (CharlieKiosk v2)

- We did it!!!!
- Screen isn't upside down
- Mounting ports on the side
- Sleek all black design
- Custom internal mounting :)



March 2023

Vending Machine v6 (CharlieKiosk v2)

- We did it!!!!
- Screen isn't upside down
- Mounting ports on the side
- Sleek all black design
- Custom internal mounting :)

I broke it
(again)



March 2023

Ingredients

- Raspberry Pi 4
- PN532 NFC reader
- Raspberry Pi Touchscreen
- Adafruit Enclosed Speaker Set
- PLA chassis + bracket
- Acrylic front + back panels
- Usb-c port
- On/off switch
- Blood
- Sweat
- Tears

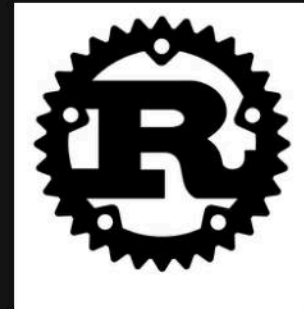
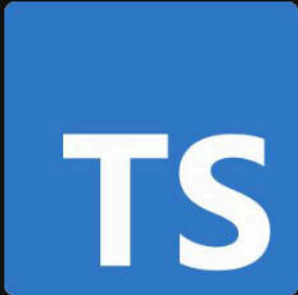


Then One Day



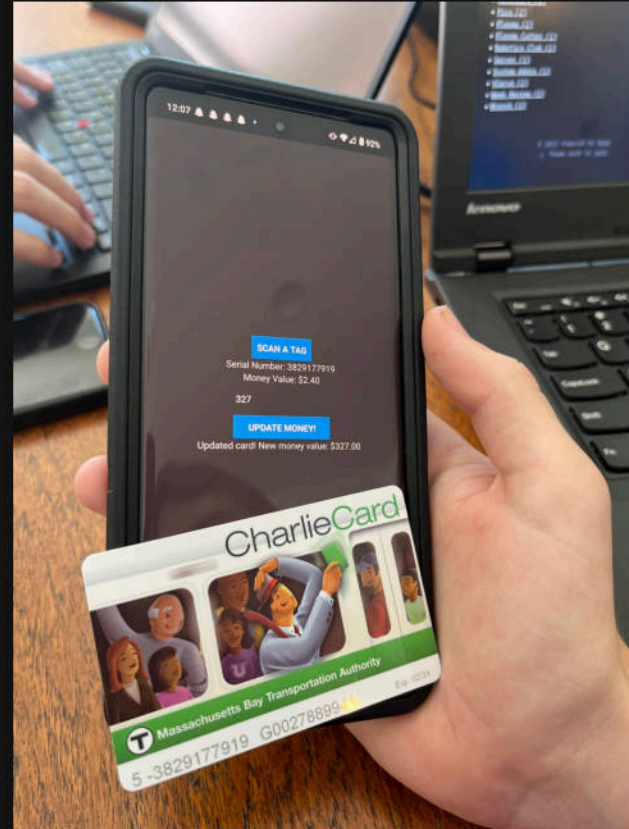
Software

- Vending machine UI made using Tauri
- Charlie Forge library on the backend
- NFC connection on a Rust backend
- Solid JS frontend
- Users can update date, money, expiration date, and card type



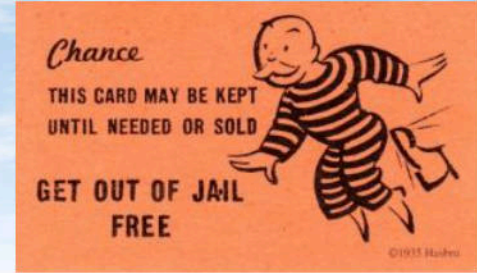
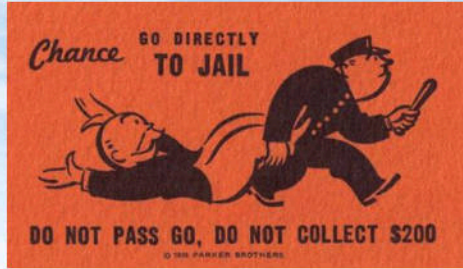
Android App

- Slapped together in 2 nights
- React Native
- Ugly



February 2023

Part 5: Not Ending Up In Handcuffs



Your CharlieCard can be hacked by an Android phone, MBTA admits

By [Hiawatha Bray](#) Globe Staff, Updated December 8, 2022, 1:28 p.m.



A CharlieCard being used at the State Street MBTA station in Boston. JOHN BLANDING/GLOBE STAFF



Bobbyr

Dec 8, 2022 · 18 min read · [Listen](#)



Operation Charlie: Hacking the MBTA CharlieCard from 2008 to Present



Photo by Garrett Quinn

December 8, 2022



Contact

First Name

Last Name

Email

Write a message

Submit



Reaching out to the T

Due 1.19: In Our Time Chap...

PowerManagement - Ubuntu...

Customer Support | MBTA

Privacy Policy | Policies | M...

Operation Charlie: Hacking

20 Entertaining Uses of Chat

CharlieCard - Wikipedia

mbta.com/customer-support

How To Use Ar...

Customer Support

Email Us

You can expect a response to most tickets within 5 business days. Accessibility complaints require a full investigation, which may take up to 30 days.

All fields with an asterisk* are required.

Message

Category*

☐ Complaint

☐ Comment

☒ Question

☐ Compliment

Subject*

Other

Let us know how we can help*

Hi there,
My name is Matthew Harris and I'm a junior at Medford Vocational Technical High School. A group of students and I have been conducting research on the security of the CharlieCard. We believe we have found a vulnerability and would like to know if there is a place where we can responsibly disclose this information.

Regards,
Matthew Harris

Call Us

Monday - Friday: 6:30 AM - 8 PM
Saturday - Sunday: 8 AM - 4 PM

Main Hotline: 617-222-3200
Toll Free: 800-392-6100
TTY: 617-222-5146

Elevator/Escalator Hotline: 617-222-2828

Lost and Found

Follow the link below to find the number associated with the mode or route where you lost your item.

[View phone numbers](#)

Get Service Updates

Receive notifications of MBTA service alerts by email or text.

[Sign up for T-Alerts](#)

[Follow @MBTA on Twitter](#)

January 2023

We Were Expecting This



And they responded!



Good afternoon, Matthew

Thank you for outreaching to us here at the MBTA and we do have a Vulnerability Disclosure Program (or VDP).

As part of the process, we will share terms for you to safely disclose, support us to understand the issue, safe-harbor provisions for you, and ultimately the timeline/terms so that you can publicly disclose or publish.

I'll work with our team to get this over to you in the next day.

In the interim, wanted to see if you are aware that we recently supported a similar case which was in the Boston Globe and local news regarding a CharlieCard vulnerabilities.

<https://www.bostonglobe.com/2022/12/08/business/your-charliecard-can-be-hacked-by-an-android-phone-mbta-admits/>

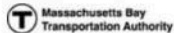
This was rather recent and not sure if you are aware. In any case, please let me know if you have any questions and in the interim, I'll get the terms to you.

Thank you again,

-Scott

Scott Margolis | Chief Information Security Officer |

| [MBTA.com](https://www.mbta.com)



They Invited Us To Their Headquarters

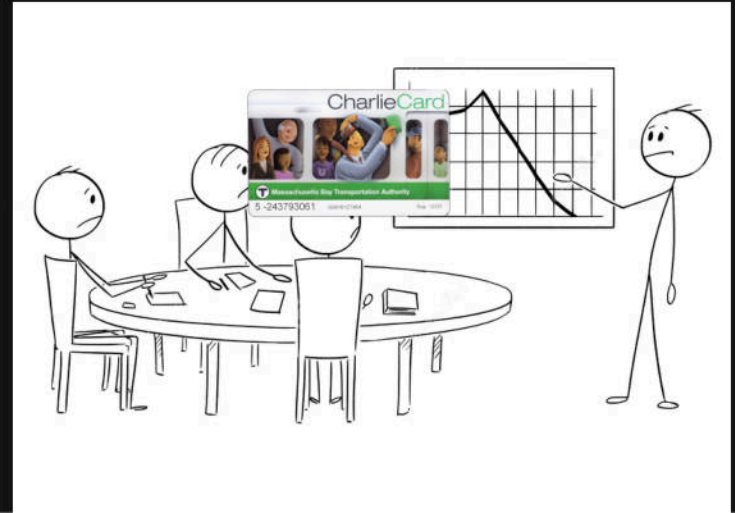


The Meeting

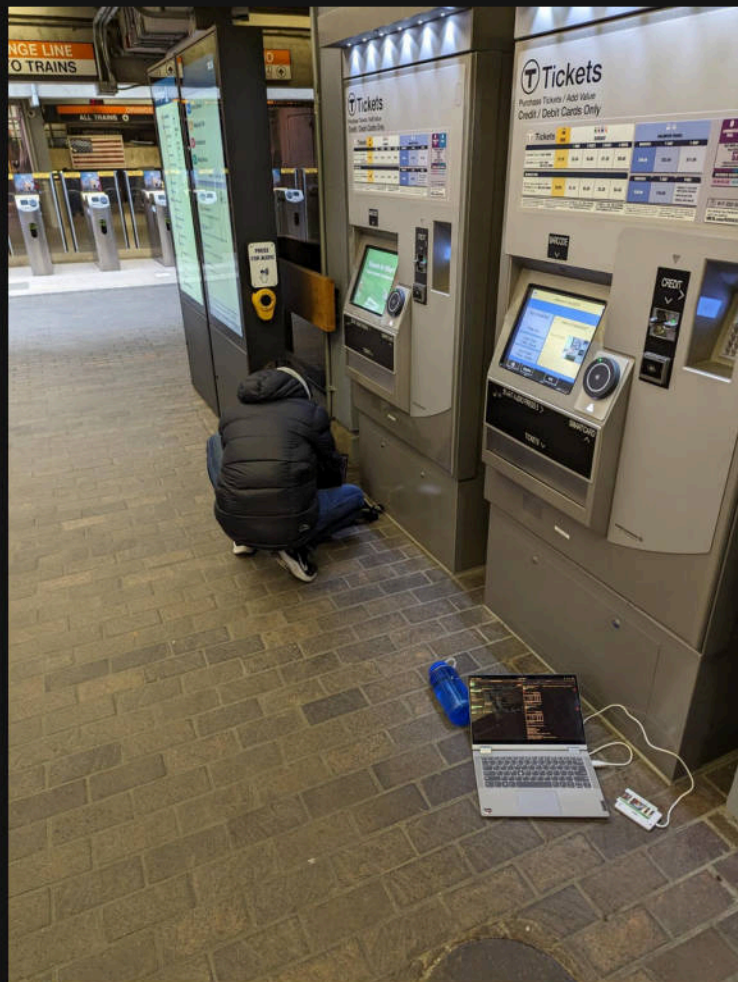
- We rolled up
- It wasn't the address to train jail
- We sat down and gave them a more professional presentation
- They were very nice and talked to us about possible solutions



February 9, 2023



Post Meeting Fun



February 2023

We Were Very Slightly Suspicious



But Nobody Stopped Us...

But Nobody Stopped Us...
Until They Did



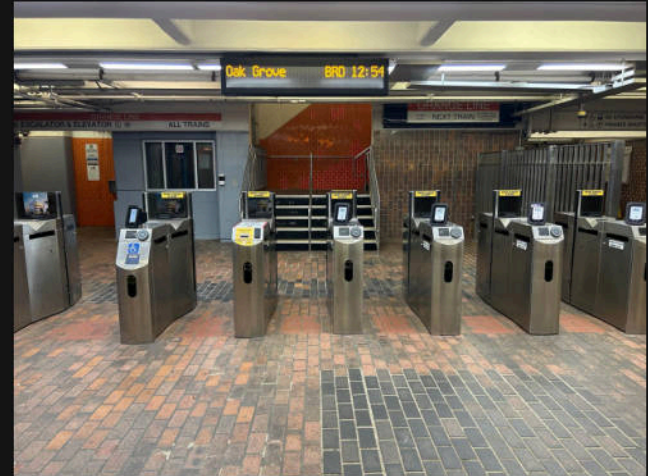
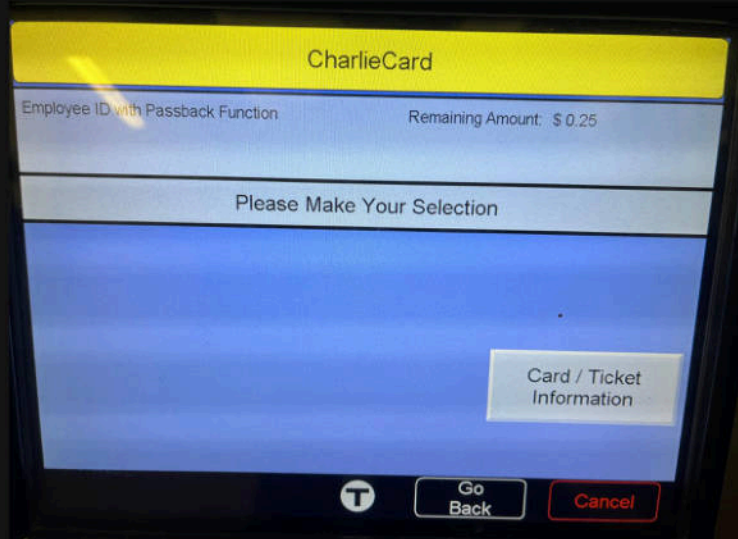
At 1AM

Take Two



February 2023

Post - Post Meeting Fun



March 2023

Demo Time



3 days ago

Recap and Takeaways

Recap

- **We reverse engineered the MBTA's CharlieCards**
- **Gave ourselves free rides for life (or until they fix their system)**
- **Reported the vulnerability to the MBTA and met with their security team**
- **Created our own fare vending machine and an Android app**

Takeaways If You're A Hacker

- Free rides if you come to Boston
- Everything's more fun with friends
- Reverse engineering strategies
- Reaching out to a government agency strategies



Things to Keep In Mind

- The same pitfalls should be avoided when the next system is rolled out
 - **Don't store money on the card**
 - Make use of pre-existing secure systems such as Apple and Google Pay
- It's more secure to store cards, transactions, and users in a database
- Storing in a database also means that detecting vulnerabilities will be easier
- Databases also make studying analytics easier
- **Systems need to be maintained after they are made!**

Acknowledgments

Finn Sedan



- Pitched in money for our first NFC reader
- Drove us around
- Cool guy

Bobby Rauch



- Helped us get in contact with the MBTA
- All around helpful guy

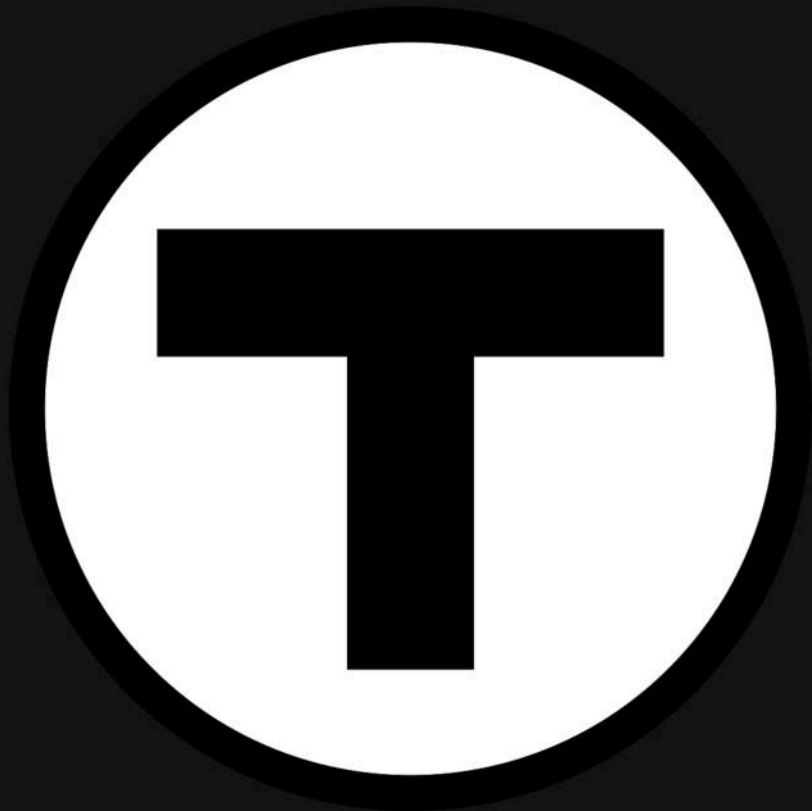
Mr. Christy



Mrs. Miller



MBTA



- Was very cooperative throughout the whole process
- Did not sue us
- Everyone was very nice to us
- And remember:

"The MBTA does not endorse or encourage this type of conduct. evading fares and/or hacking the MBTA fare system is illegal and the MBTA takes these matters seriously. Anyone caught engaging in this type of behavior will be referred to the appropriate law enforcement agency. The MBTA has implemented mitigation measures to address some of these concerns."

If you wanna contact us

Matthew Harris - mharris06@protonmail.ch, mattyharris.net

Scott Campbell - joseph_scott_campbell@protonmail.com,
josephscottcampbell.com

Zachary Bertocchi - zbertocchi@gmail.com, zackbertocchi.com

Noah Gibson - thebottemofthebarrel@gmail.com, noah.nopreserveroot.xyz

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<https://www.cpusa.org/wp-content/uploads/2022/08/FBI-swat-team.jpg>

<https://www.bostonglobe.com/2022/12/08/business/your-charliecard-can-be-hacked-by-an-android-phone-mbta-admits/>

<https://medium.com/@bobbyrsec/operation-charlie-hacking-the-mbta-charliecard-from-2008-to-present-24ea9f0aaa38>

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<https://cdn.mbta.com/sites/default/files/fares/charliecard-tap-farebox.jpg>

<https://github.com/rust-lang/rust-artwork/blob/master/logo/rust-logo-512x512-blk.png>

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Thanks for Watching!

